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Chapter
Probability

1. A bag contains 8 green and 10 white balls. Two balls are drawn. What is the probability that one is green and other is white?**Ans:** $\frac{80}{153}$
2. A bag contains 6 white, 4 red and 10 black balls. Two balls are drawn at random. Find the probability that they will both be black.**Ans:** $\frac{9}{38}$
3. Two balls are drawn from a bag containing 8 green and 10 white balls. what is the probability of drawing balls of a) same color b) different color**Ans:** $\frac{73}{153}, \frac{80}{153}$
4. A bag contains 9 red, 7 white and 4 black balls. A ball is drawn at random. What is the probability o drawing a) a red ball b) a white ball c) not a black ball?**Ans:** $\frac{9}{20}, \frac{7}{20}, \frac{16}{20}$
5. A consignment of 15 pens contains 2 defective pens. Two pens are taken at random from the consignment without replacement. what is the probability that **Ans:0.00952, 0.7428 PU2005**
   a. both are defective
   b. both are non-defective
6. A bag contains 3 red, 4 white and 5 black balls. Three balls are drawn from the bag. Find t probability that
   a. all are black**Ans:1/22, 3/11, 3/44**
   b. all are different color
   c. All are same color.

7. Three bags contain 3 video cards and 2 network cards, 5 video cards and 6 network cards, 2 vid cards and 4 network cards respectively. One card is drawn from each urn. Find the expected numb of video cards **PU2018**
8. There are 3 economist, 4 engineers, 2 statisticians and 1 doctor. A committee of 4 from among them to be formed. find the probability that the committee **Ans:4/35, 5/6, 2/5**
   a. consists one of each kind
   b. has at least one economist
   c. has the doctor as a member and three others
9. From a group of 3 Nepalese, 4 Indian and 5 Americans, a subcommittee of four persons are selec randomly. Find the probability that the subcommittee will consists of **PU2016**
   a. 3 Nepalese and 1 Indian
   b. 1 Nepalese, 1 Indian and 2 Americans
   c. 4 Americans**Ans: 4/495, 4/33, 1/99**
10. In a company, there are 4 civil engineers, 5 IT officers and 6 finance offi

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13. Five men in a company of 20 are graduates. If 3 men are picked out of the 20 at random What is the probability that they are
    a. All graduates
    b. 2 graduates
    c. At most 1 graduates **PU2015 Ans:1/114**
14. A bag contains 20 balls numbered from 1 to 20. One ball is drawn at random from the bag. find the probability that the drawn balls are multiple of a) 3 or 5 b) 5 or 7**Ans:9/20, 3/10**
15. A card is drawn from a bag containing 10 cards numbered 1 to 10. Let A denotes the event that the number drawn in the card is less than 7. B is the event that the number on the card is multiple of 2. Find the probabilities of the following events. **PU 2003**
    a) A^c b) B c) $(A \cap B^c)$ d) AUB e) $A^c \cap B$ f) B/A
16. A card is drawn at random from a well shuffled pack of 52 cards, find the probability of getting
    a) a jack, a queen or a king **Ans:** $\frac{3}{13}$
    b) a diamond or a king **Ans:** $\frac{4}{13}$
17. The HAL Corporation wishes to improve the resistance of its personal computer to disk driver and keyboard failure. At present, the design of the computer is such that disk driver failure occur only one third as often as keyboard failures. The probability of simultaneous disk drive and keyboard failure is 0.05.
    a) If the computer is 80% resistance to disk drive or keyboard failure, how low must be the disk drive failure probability be?
    b) if the keyboard is improved so that it fails only twice as often as the disk drive (and the simultaneous failure probability is still 0.05) will disk drive failure probability from part a) yield a resistance to disk drive or keyboard failure highest or lower than 90%?
    Ans: 0.0625, 86.25% which is lower PU2015
18. Mr A and B both are interested to attend a seminar in Department of management, PU. The chance of attending a seminar by Mr A is 0.6 and that by Mr B is 0.3. They both can also attend the seminar. what is the probability that
    a. At least any one of them will attend the seminar?
    b. Only one of them will attend the seminar? **Ans:0.72,0.54,0.28**
    c. None of them will attend the seminar?
19. The probability that a new airport will get an award for its design is 0.16, the probability that it will get award for the efficient use of material is 0.24, and the probability that it will get both awards is 0.11.
    a. What is the probability that it will get at least one award?
    b. What is the probability that it will get only one award?**Ans:029,0.18,.458**
    c. If it will get an award for the efficient use of material then what is the probability that it will also get an award for its design?
20. The probability that a student passes in economics test is ¾ and the probability that he passes both economics and management test is 12/40. The probability that he passes at least one test is 5/6. Find the probability that he will pass in the management test. **Ans:23/60**
21. The probability that an integrated circuit chip will have defecting etching is 0.12, the probability that it will have crack defect is 0.29 and the probability that it has both defects is 0.07. **Ans:0.34,0.66**
    a. What is the probability that a newly manufactured chip will have at least one of two defects?
    b. What is the probability that a newly manufactured chip will have neither any kind of defect?
22. In a certain locality of town, a survey of 600 women about the fuels for cooking revealed that 230 use kerosene oil, 175 women use electricity and 40 women use both. Find the probability that a women selected use,
    a) At least one of the fuel?
    b) Neither of the fuel?
    c) Kerosene oil only?
    d) Electricity only?
    e) Exactly one?

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55. A random variable X has following probability function

Value of X	-2	-1	0	1	2	3
P(x)	0.1	k	0.2	2k	0.3	k

a. find the value of k
b. find P(X<2), P(-2<X<2), P(0≤X≤2) Ans: k=0.1, 0.6, 0.5, 0.7, 0.1, 0.4, 1
c. find F(-2), F(0) and F(3)
d. construct the cdf F(x)

56. if a discrete random variable has the probability function as

X	0	1	2	3	4	5	6	7	8
P(x)	k	2k	3k	4k	5k	4k	3k	2k	K

a. determine the value of k
b. find P(X<4), P(X≥4), P(0<X<5) **Ans: 1/25, 10/25, 15/25, 14/25**
c. find cdf of X

57. If $P(X=x)=\frac{x}{15}$, $x = 1,2,3,4,5$
0, otherwise

Find a) P(X=1 or 2) and b) $P(\frac{1}{2} < X < \frac{5}{2}/X>1)$ **Ans: 1/5, 1/7**

58. Given the pdf **Ans: 6,0.2844, 0.608**

$f(x)= kx(1-x)$, $0<x<1$
0 otherwise

Find a) k b) P (0.3<X<0.5) c) P(X<0.4)

59. A random variable X has the density function $f(x)= a+bx^2$, $0<x<1$.
Determine the value of 'a' and 'b' such that a+b=2 **Ans: 0.5, 1.5**

60. Suppose that X is a random variable, whose probability density function is given by, **PU201**

$f(x)= C(4x-2x^2)$ $0<x<2$
0 otherwise

a. determine the value of C
b. determine the cumulative distribution function of X **Ans:3/8, $x^2(3-x)/4$, 0.5**
c. find P(X>1)

61. The probability distribution for the number of defective items (x) in a sample of 4 is as follows: **Ans:0.99,0.869,0.932**

X	0	1	2	3	4
P(x)	0.35	0.39	0.19	0.06	0.01

Calculate the expected value, variance and standard deviation for x. **PU20**

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62. A r.v. X has following probability distribution. **PU2019**

X	2	3	4	5	6	7
P(x)	0.1	k	0.2	2k	0.3	k

a) Find the value of k
b) Find E(X) and V(X)
c) Find the distribution function of X

63. A continuous random variable X has the following density function. **PU2018**

$$f(x) = kx^2 \quad for\ 0 < x < 1$$
$$0 \quad otherwise$$

I) Find the value of k
II) P(0<X<0.7)
III) E(X)

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64. A random variable X has the following probability function

X	0	1	2	3	4	5	6	7
P(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find

a) k
b) $P(X \geq 6)$
c) $P(X<6)$ **Ans: 0.1, 4/5, 1/5, 0.19. 3.66 3.41**
d) $P(0<X<5)$
e) mean and variance of X

65. A random variable X has the following probability function.

X	2	3	4	5	6	7
P(x)	0.1	k	0.2	2k	0.3	k

Find the value of k and calculate mean and variance. **Ans: 0.1, 4.8, 2.16 PU2017**

66. An appliance dealer sells different models of freezers having 13.5, 15.9 and 19.1 cu. Ft. of storage space purchased by the next customer to buy a freezer and p(x) be the pmf given below as:**PU2016**

x	13.5	15.9	19.1
p(x)	0.2	0.5	0.3

a. compute E(X) and V(X) **Ans: 16.38 3.994 401 2496.25**
b. If the price of a freezer having capacity x cubic feet is 25X-8.5, what is the expected price paid by the next customer to buy a freezer?
c. What is the variance of the price 25X-8.5 paid by the next customer?

67. A random variable X had the following probability function. **PU2003**

X	0	1	2	3	4	5	6	7	8
P(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

Find

a. the value of a **Ans: 1/81, 1/9, 24/81**
b. $P(X<3)$ and $P(0<X<5)$
c. the mean and variance of X
d. $P(X \leq 3)$, $P(0<X<5)$ and F(7) **Ans: 61/81, 24/81, 64/81**
e. Find out the distribution function of a random variable X?

68. A random variable X has the following probability distribution. **Ans: 0.1, 0.8, 2.16, 6.6, 8.64**

Values of X	-2	-1	0	1	2	3
P(x)	0.1	k	0.2	2k	0.3	k

Find the value of k and calculate mean and standard deviation of the distribution. Also find E(2X+5) and V(2X+5)

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69. Two balanced dice are thrown. Find the expected value of the sum of numbers of points on them. Also find variance.**PU2009 Ans: 7, 5.83**

70. A random variable X is defined to be the larger of the two values when two dice are thrown or the value if the values are the same.

Find;

a) E(10X+5)
b) $E(X^2)$ **Ans: 49.7, 21.97, 1.41, 31.54%**
c) SD of X
d) coefficient of variation of X

71. In a business venture a man can make a profit of Rs. 50,000 or incur a loss of Rs. 20,000. The probability of making profit or incurring loss from the past experience is known to be 0.75 and 0.25 respectively. What is his expected profit and variance?Ans: 32,500 9.18×10^9

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72. A random variable X is defined to be the difference between the higher value and the lower value when two dice are thrown. If they have the same value, X is defined to be zero. Find Ans: 13.88, 5.83, 1.44, 74.23% .
    a) E(2X+10)
    b) $E(X^2)$
    c) SD of X
    d) CV of X
73. The following table presents a discrete probability distribution associated with the daily demand a product.

Number demand per day (X)	10	20	30	40	50	Total
P(X)	0.08	0.24	0.28	0.3	0.1	1

    a. Determine the mean daily demand
    b. What is the standard deviation of daily demand? **Ans: 31, 11.18**
74. A coin is tossed until a head appears. What is expectation of number of tosses required? **PU2013 Ans:2**
75. On an average how many times must a dice thrown until one get a 6? **PU2015Ans: 6**
76. From a lot of 10 items containing 3 defectives a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. Answer the following.
    a. E(X)
    b. V(X)**Ans: 1.2 0.56PU2016**
    c. The probability distribution of X
77. An urn contains 7 red and 4 white balls, 3 balls are drawn at random. Find the probabil distribution of number of white balls drawn. Also find the mathematical expectation and variance the distribution. **PU2012 Ans:1.09 0.55**
78. A bag contains 5 white and 3 red balls. Three balls are taken at random. Find the expected number white balls drawn in the three. **Ans:1.875**
79. A bag contains 4 white and 6 red balls. Three balls are drawn randomly from the bag. If each wh ball drawn carries a Rs. 4 and red ball Rs. 6. Find the expected amount received. **Ans: 15.6**
80. From a lot containing 50 items, 10 of which are defective, 4 are chosen at random. Obtain probability distribution of the number of defective items drawn. Also find the probability of 5 d defective items. **PU2003**
81. The length of time (in minutes) that a certain lady speaks on the telephone is found to be rand phenomenon, with a probability function specified by the probability density function f as:**PU2015**

$$f(x)=Ae^{-x/5} \quad x\geq 0$$
$$0 \quad \text{otherwise}$$

    a. find the value of A**Ans:1/5, e^{-2} 5 25 e^{-1}, e^{-2}-e^{-1}**
    b. What is the probability that the number of minutes that she will talk over the phone is m than 10 minutes?
    c. What is the probability that she will talk over the phone is
        I) More than 10 minutes
        II) Less than 5 minutes
        III) Between 5 and 10 minutes
    d. Find the mean and variance of talk time on the telephone.
82. A random variable X has the density function $f(x)= a+bx^2$, $0<x<1$. Determine a and b so that m is $\frac{2}{3}$. Also find the variance of the distribution.**Ans: 1/3 , 2 , 3/45**
83. A random variable X is distributed at random between the values 0 and 1 so that its pdf is $f(x)=kx^3(4-x)^2$, $0<x<1$ **Ans: 30/77, 422/539 and 0.0295**

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Where, k is a constant. Find the value of k, using this value of k find mean and variance of distribution.

84. A random variable X has pdf as

$f(x)=kx(2-x)$, $0\le x\le 2$

0, otherwise

Find the value of k, mean and variance of random variable. **Ans:3/4, 1, 1/5**

85. Let X be a continuous random variable with pdf

$f(x)=ax$, $0\le x\le 2$

$=a$, $2\le x\le 4$

$=ax+3a$, $4\le x\le 6$

a) Determine the constant a and find
b) $P(X\le 1.5)$
c) $P(1.5\le X\le 2.5)$ **Ans: 1/20, 0.056, 11/160, 2.34**
d) Obtain E(x)

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23. The chance that A can solve a certain problem is $\frac{2}{3}$, the chance that B can solve it is $\frac{3}{4}$. if they both try, find the probability that
    a) A solves it but B cannot?
    b) B solves it but A cannot?
    c) Both of them cannot solve it?Ans: $\frac{1}{6}, \frac{1}{4}, \frac{1}{12}, \frac{11}{12}, \frac{5}{12}$
    d) At least one of them will solve it?
    e) Only one of them solves it?
24. A husband and wife appear in an interview for the same post. The probability of husband's selection is $\frac{1}{7}$ and that of wife's selection is $\frac{1}{5}$. What is the probability a) both of them will be selected b) none of them will be selected c) at least one of them will be selected d) only one of them is selected?**Ans:** $\frac{1}{35}, \frac{24}{35}, \frac{11}{35}, \frac{2}{7}$
25. The probability that a boy passes the examination is $\frac{3}{4}$ and that of a girl is $\frac{2}{3}$. Find the probability that a) both will pass the examination b) at least one of them will pass the examination.**Ans:** $\frac{1}{2}, \frac{7}{6}$
26. A can hit a target 4 times in 5 shots. B can hit the target 3 times in 4 shots and C can hit the target 2 times in 3 shots. They fire a volley. What is the probability that the target is damaged if 2 hits are needed for it?**Ans:0.83**
27. A problem of statistics is given to three students A, B and C where chance of solving it is 1/2, 3/4 and 2/3 respectively. If they independently solve it, what is the probability that
    i) All can solve the problem
    ii) None can solve the problem
    iii) Problem can be solved**Ans:** 0.25, 0.042, 0.958, 0.25
    iv) Exactly one can solve the problem
28. A problem of statistics is given to three students A, B and C where chance of solving it is 1/3, 1/4 and 1/5 respectively. If they independently solve it, what is the probability that
    i) All can solve the problem
    ii) None can solve the problem
    iii) Problem can be solved**Ans:** 0.017, 0.2, 0.6, and 0.433 **PU2017**
    iv) Exactly one can solve the problem
29. An oil exploration company currently has two active projects, one in Asia and the other in Europe. Let A be the event that the Asian project is successful and B be the event that the European project successful. Suppose that A and B are independent events with P(A)=0.4 and P(B)=0.7**PU2016**
    a. If the Asian project is not successful, what is the probability that the European project is also no successful? Explain your reasoning.**Ans:**0.18
    b. What is the probability that at least one of the two projects will be successful?**Ans:0.82**
    c. Given that at least one of the two projects is successful, what is the probability that only the Asian project is successful?**Ans:0.12**
30. A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is 1/7 and that of wife's selection is 1/5. what is the probability that
    a) Both of them will be selected?
    b) Only one of them will be selected? Ans:1/35, 2/7, 24/35
    c) None of them will be selected?

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Condition &Baye's Theorem Problems

31. In a certain University of Nepal 80% of the students are Hindu and remaining non Hindu. If 20% of the Hindu students are smokers while 25% of the non Hindu are smokers. What is the probability that a randomly selected smoker students is a non Hindu? **PU2015** **Ans:0.238**

32. Nepal Telecom announced an advertisement for the post of telecommunication engineer to fulfill one vacant post for its Butwal branch. The application of the potential candidates is prepared in the following table.

Age (years)	Bachelor in engineering	Master of engineering
Under 25	60	8
25-30	7	26
Above 30	9	10

If one of the applications is selected at random, find the probabilities that **Ans: 0.526, 0.0583**

a) He has completed master in engineering given that he is above 30 years old.
b) He has done bachelor in engineering and lies between 25 to 30 years.

33. A factory has three machines A, B and C producing large number of a certain item if the daily production of the item 50% are produced on A, 30% on B and 20% on C. Record show that 2% of the item produced on A are defective, 3% of items produced on B are defective and 4% of items produced on C are defective. The occurrence of a defective item is independent of all other items. One item is chosen at random from a day's total output.
   a. Show that the probability of it being defective item is 0.027.
   b. Given that it is defective, find the probability that is was produced on machine A.**Ans:0.3703**

34. In a bolt factory machine A, B and C manufactures respectively 25%, 35% and 40%. of the total of their output 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. what is the probability that it was manufactured by machnie A? **Ans: 0.36**

35. There are three machines A, B and C producing 1000, 2000 and 3000 articles per hour respectively. These machines are known to be producing 1%, 2% and 3% defectives respectively. One article is selected at random from an hour production of the three machines and found to be defective. What is the probability that the article is produced from i. machine A ii. Machine B iii. Machine C? **Ans:0.071,0.286,0.643**

36. A firm has 80% of its service calls made by a contractor and 10% of these calls result in customer complaints. The other 20% of the service calls are made by their own employees, and these calls have a 5% complaint rate. Find the probability of a complaint. Also, using Bayes theorem to find the probability that a complaint was from a customer whose service was provided by the contractor. **PU2018**

37. A company has two plants to manufacture scooters. Plant I manufacturers 80% of the scooters and plant II manufactures 20%. At plant I, 85 out of 100 scooters are rated standard quality or better. At plant II, only 65 out of 100 scooters are rated standard quality or better. What is the probability that scooter selected at random came from plant I if it is known that the scooter is of standard quality?**Ans: 0.84**

38. Customer are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products. **PU2019**
   a) What is the probability that a product attains a good review?
   b) If a new design attains a good review, what is the probability that it will be a highly successful product?

39. The probabilities of the three persons A, B and C becoming VC of PU are $\frac{3}{9}, \frac{4}{9}$ *and* $\frac{2}{9}$ respectively. The probabilities that the grade increment system will be introduced if A, B and C becomes VC are $\frac{1}{7}, \frac{2}{7}$ *and* $\frac{3}{7}$ respectively. If the grade increment system has been introduced. **Ans : 0.47, 0.176**
   a. What is the probability that the VC appointed is Mr. B?

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b. What is the probability that the VC appointed is Mr. A?

40. In a certain factory machines I, II and III are all producing springs of the same length. of their production machines I, II and III produces 2%, 1% and 3% defective springs respectively. of the total productionof springs in the factory, machines I produces 35%, machines II produces 25% and machine III produces 40%. if one spring is selected at random from the total springs produced in a day, find **Ans: 0.0215, 0.5581**
    a. the probability that it is defective
    b. the conditional probability that it was produced by machine III.
41. An insurance company insured 3000 scooters, 4000 cars and 5000 trucks. The probability that an accident occurred due to scooter, car and truck is 0.02, 0.03 and 0.04 respectively. One of the insured vehicle meet with an accident. Find the probability that it is a car. **Ans:6/19**
42. The chances of X, Y, Z becoming manager of a certain company are 4:2:3. The probabilities tha bonus scheme will be introduced if X, Y, Z becomes manager are 0.3, 0.5 and 0.8 respectively. If th bonus scheme has been introduced, what is the probability that X is appointed as the manager **Ans:0.26**
43. Two production plants A and B make wire cables that are sent to a common distributor. 40% of the cables sent to the distributors come from Plant A and the remaining 60% from Plant B. Among the cables produced at Plant A 95% meet the strength specifications. Among the cables produced at Plar B 98% meet the strength specification. **PU2015**
    the distributor selects one cable at random from among all cables in stock. if the cable select is found to meet the strength specifications, what is the probability that the cable was produced at Plant A? **Ans:0.395**
44. A manufacturing firm produces steel pipes in three plants with daily production volume of 500, 100 and 2000 units respectively. According to the past experience, it is known that the fraction of defective outputs produced by the three plants are respectively 0.005, 0.008 and 0.01. if pipe is selected from a day's total production and found to be defective. From which plant the defective pip is expected to have been produced?**PU2016**
45. Three road construction firms X, Y and Z bid for a certain contract. from past experience, it is estimated that the probability that X will be awarded the contract is 0.4, while for Y and Z the probabilities are 0.35 and 0.25 respectively. if X does receive the contract, the probablity that the work will be satisfactorily completed on time is 0.75. for Y and Z these probabilities are 0.8 and 0.7
    a. what is the probability that the work will be completed satisfactorily?**Ans:0.755,0.3709**
    b. it turns out that the work was done satisfactorily. what is the probability that Y was awarded the contract?**PU2016**
46. An insurance company insured 2000 Civil Engineers, 4000 Electrical engineers and 6000 Mechanic engineers working in Hydropower sector. The probability of an accident involving Civil engineer, Electrical engineer and Mechanical engineer during their Jobs are 0.01, 0.03 and 0.15 respectively.on of the insured engineer meets with an accident. What is the probability that he is Civil engineer?
47. In a class of 75 students, 15 students were considered to be very intelligent, 45 as medium and re below the average. The probability that a very intelligent student fails in an examination is 0.005, t medium student failing has probability 0.05 and corresponding probability for a below average 0.15. If a student is known to have passed the examination, what is the probability that he is belo the average? **PU2013 Ans:0.181**
48. The contents of Urns I, II and III are as follows: **PU2009**
    1 white, 2 black and 3 red balls
    2 white, 1 black and 1 red bail***Ans: 0.466 0.2797 0.254***
    4 white, 5 black and 3 red balls
    one urn is chosen at random and two balls are drawn. they happen to be white and red. what is the probability that they come form Urn I , II and III?
49. A physical therapist working at an institute of Medicine, Kathmandu knows that the football tea will play 40% of its games on artificial turf this season. He also knows that a football playe chances of incurring a knee injury are 50% higher if he is playing on artificial turf instead of grass.

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a player's probability of knee injury on an artificial turf is 0.42, what is the probability thatAns:0.336, 0.5

a. A randomly selected football player incurs a knee injury?

b. A randomly selected football player with a knee injury incurred the injury playing on grass?

50 An advertising executive is studying television viewing habits of married men and women during prime time hours. On the basis of post viewing records, the executive has determined that during prime time husband are watching television 60% of the time when the husband is watching television 40% of the time the wife is also watching when the husband is not watching television 30% of the time the wife is watching it. Find the probability that**Ans:0.36,0.67**

c) The wife is watching television in prime time.

d) If the wife is watching television, the husband is also watching television.

51. An oil explorer orders seismic tests to determine whether oil is likely to be found in a certain drilling area. The seismic tests have a known reliability. When oil does exist in the testing area, the test will indicate so 85% of the time; when oil does not exist in the test area, 10% of the time the test will erroneously indicate that it does exist. The explorer believes that the probability of existence of an oil deposit in the test area is 0.4. If test is conducted and indicates the presence of oil, what is the probability that an oil deposit really exists? Ans:0.85

52. . Mr. KC has been teaching Probability and Statistics for many years. He knows that 80% of the students will complete the assignment. He has also determined that among those who do their assignment, 90% will pass the course. Among those students who do not do their assignments, 60% will pass. Bishal took Statistics last semester from Mr KC and has passed. What is the probability that he has completed the assignment? Ans:0.857

53. A student knew only 60% of the questions in a test each with 5 answers. He simply guessed while answering the test. What is the probability that he knew the answer to a question given that he answered it correctly?**PU2014** **Ans:0.88**

54. In a certain recruitment test, there are multiple choice questions. There are 4 possible answers to each question and of which one is correct. An intelligent student knows 80% of the answer. If intelligent student gets the correct answer, what is the probability that he is guessing? **PU2015** **Ans:0.0588**

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Binomial Distribution

86. Ten coins are tossed simultaneously. Find the probability of obtaining
    a. No head
    b. Exactly 6 heads
    c. At least one head **Ans: 0.00098, 0.2051, 0.999, 0.172**
    d. Not more than three heads
87. If the probability of recovery from a certain disease is 0.2 and 10 people came down with the disease, what is the probability that **Ans: 0.1073, 0.8926, 0.6778**
    a. None of them will recover?
    b At least one will recover?
    c. At most two will recover?
88. Test for impurities commonly found drinking water from private wells showed that 30% of all wel in a particular have impurity A. if a random sample of 5 wells is selected from the large number of wells in the country, what is the probability that: **Ans:0.1323, 0.16308, 0.83692**
    a. Exactly 3 wells have impurity A?
    b. At least 3 wells have impurity A?
    c. Fewer than 3 wells have impurity A?
89. In a sampling of 10 items, there is a chance of getting a defective item is 40%. Find the following probabilities.
    a. There are exactly two defective items in the sample.
    b. There are at least three defective items in the sample.
    c. There are at most seven defective items in the sample
90. 45% of the Nepalese workers have been gone abroad are illegal. If in a sample of six, Nepalese workers who have gone abroad, what is the probability that
    i. 3 are illegal
    ii. All are legal. **Ans: 0.3032, 0.0277, 0.9917**
    iii. At least one is legal
91. The latest nationwide political poll indicates that for American, who is randomly selected, the probability that they are conservative is 0.6 and the probability that they are liberal is 0.4. Assumi that these probabilities are accurate, answer the following questions pertaining to a randomly cho group of 10 Americans.
    a. What is the probability that 4 are liberal?
    b. What is the probability that none are conservative? **Ans: 0.2508, 0.0001**
92. A multiple-choice test has 5 questions. There are 4 choices for each question. A student who has studied for the test decides to answer all questions randomly. What is the probability that he will
    a. Five questions correct? **Ans: 0.001, 0.0156**
    b. At least four questions correct?
93. Multiple choice tests consist of 10 questions and 4 answers to each question. If each question is answered by shuffling 4 tags labeled 1, 2, 3 and 4 drawing one and making the alternative whose number is drawn. Find the probability of a) getting 3 b) at least one c) at least 7 d) at most 3 of the questions answered correctly.
94. Harley Davidson, director of quality control for the Kyoto Motor company, is conducting his monthly spot check of automatic transmission. In this procedure, 10 transmissions are removed fr the pool of components and are checked for manufacturing defects. Historically, only 2% of the transmissions have such flaws. (Assume that flaws occur independently in different transmission
    a. What is the probability that Harley's sample contains more than two transmissions with manufacturing flaws? **Ans:0.0009, 0.8171 PU2015**
    b. What is the probability that none of the selected transmission has any manufacturing flaw
95. If the mean of the binomial distribution is 0.4 and its standard deviation is 0.6, find the probabilit at least one success. **Ans: 0.3439**
96. In a binomial distribution with 6 independent trials the probabilities of 3 and 4 successes are foun to be 0.2457 and 0.0819. Find the parameter 'p' of the binomial distribution. **Ans:4/1**

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97. Mean and variance of binomial distribution are 6 and 4 respectively find n, p and q.

98. Mr Lucia is the mayor of a large city. Lately, she has become concerned about the possibility that large numbers of people who are drawing unemployment checks are secretly employed. Her assistants estimate that 35% of unemployment beneficiaries fall into this categories but Ms. Ruffner is not convinced. She asks one of her aides to conduct a quiet investigation of 8 randomly selected unemployment beneficiaries.
    a. If the mayor's assistants are correct, what is the probability that more than six of the individuals investigated have jobs?**Ans: 0.004, 0.28**
    b. If the mayor's assistants are correct, what is the probability that only three of the investigated individuals have jobs?

99. A local hospital has been conducting an experiment program to assist person to stop cigarette smoking. Upon completion of the program, participants realize a 60% success rate. If the sample of 4 participants is selected at random, what is the probability that
    a. All four will stop smoking?
    b. One will stop smoking?
    c. One least one will have stopped smoking?

100 A recent study of how Nepalese spend their leisure time surveyed workers employed more than 5 years. They determined the probability an employee has 2 weeks of vacation time to be 0.45, one week of vacation time to be 0.1 and three or more weeks to be 0.2. Suppose 12 workers are selected at random. Answer the following questions.
    a. What is the probability that at most two of the workers have three or more weeks of vacation time?
    b. What is the probability that at least 3 workers have one week of vacation time?

101. At a particular University it has been found that 10% of the students withdraw without completing the Business Statistics course. Assume that 15 students have registered for the course this semester.
    a. What is the probability that none will withdraw?
    b. What is the probability that at least one will withdraw?**Ans: 0.018, 0.982, 0.2714**
    c. What is the probability that at most 2 will withdraw?

102. An American Society of Investors Survey found 30 % of individual investors have used a discount broker. In a random sample of 9 individuals, what is the probability that
    a. Exactly 2 of the sampled individuals have used a discount broker?
    b. At least one of them has used a discount broker?

103. A survey conducted in 2011 found about 40% of the Nepalese workers has gone aboard are illegal, if in a sample of 7 Nepalese workers who had gone aboard what is the probability that
    a) Exactly 5 are illegal
    b) At least 2 are legal**Ans: 0.0774, 0.9812**

104. A marksman firing bullets at a target and probability of hitting the target at any trial is 0.7. Find the probability that his seventh shot is his fourth hit? **PU2015 Ans:0.129654**

105. Six dice are thrown 729 times. How many times do you expected at least three dice to shown a five or six? **PU2016Ans: 233**

106. The mean and variance of binomial distribution are 3 and 2 respectively. Find the probability that the variable takes values a) less than or equal to 2 b) greater than or equal to 7 **Ans: 0.377, 0.002**

107. A commercial jet aircraft has four engines. For an aircraft in flight to land safely, at least two engines should be in working condition. Each engine has an independent reliability of 90%. What is the probability that an aircraft can land safely?

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Poisson distribution

108. Airline passengers arrive randomly and independently at the passenger screening facility at a ma international airport. The average arrival rate is 4 passengers per minute.
Ans:0.3419, 0.9804, 0.0027
   a. What is the probability that two or three passengers arrive in a minute?
   b. What is the probability that at least two passengers arrive in a minute?
   c. What is the probability that exactly one passenger will arrive in two minutes?

109. A hospital switchboard receives an average of 3 emergency calls per minute. What is the probability of receiving
   a. No call in one minute? **Ans: 0.0498, 0.00021114**
   b. At most 3 call in a 5 minute interval?

110. In a certain factory turning out razor blades there is a small chance of 1/500 for any blade to defective. The blades are supplied in a packet of 10. Use Poisson distribution to calculate approximate number of packet containing no defective, one defective, two defectives and th defectives blades in a consignment of 10,000 packets.**Ans:9802, 196, 2, 0**

111. At a petrol feeling station, one customer arrives per minute in an average. What is the probabilit that
   a. Exactly 2 customers will arrive per minute?
   b. None will arrive per minute?
   c. At least one will arrive during 2 minutes?

112. If 5% of bulbs manufactured by a company are defective. Find the probability that in a sample of 500 bulbs.
   a. Less than 5 bulbs are defective
   b. At least two bulbs are defective

113. if 3% of the bulbs manufactured by a company are defective. Find the probability that in a sampl of 200 bulbs
   a. Less than 2 bulbs are defective
   b. At least two bulbs are defective

114. If in an average, three customers arrive per minute at the bank during the lunch hour. What is th probability that in a given minute **Ans: 0.224, 0.5768**
   a. Exactly two customers will arrive?
   b. More than two customers will arrive?

115. Vehicles arrive at an intersection at the rate of 10 per minute. A traffic light cycle lasts 45 secon If the number of vehicles that arrive per cycle follows a Poisson distribution, what is the probabili that exactly 7 vehicles will arrive in a cycle?**Ans:0.1465**

116. On Monday mornings, the First National Bank has one teller window open for deposits and withdrawals. Experience has shown that the average number of arriving customers in a 4 minute interval on Monday morning is 2.8, and each teller can serve more than that number efficiently. These random arrivals at this bank on Monday mornings are Poisson distributed.
   a. What is the probability that on a Monday morning exactly six customers will arrive in a 4 minute interval?
   b. What is the probability that no one will arrive at the bank to make a deposit or withdrawal during a 4 minute interval?

117. The quality control manager of Marilyn's cookies is inspecting a batch of chocolate-chip cook that has just been baked. If the production process is in control, the mean number of chip part cookies is 6.0, what is the probability that in any particular cookie being inspected **Ans:** 0. 0.0025
   a Less than five chip part will be found?
   b. None of the chip parts will be found?

118. Vehicles arrive at the rate of two per minute. The distribution of arrivals approximates a Poisson distribution.

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a. What is the probability that no vehicles arrive in a particular minute?
b. What is the probability that at least one vehicle arrive in a particular minute?

119. The standard deviation of a Poisson distribution is 2. Find the probability that X=3.**Ans:0.1954**
120. Calculate mean and variance of a poisson variable X, if P(X=4)=P(X=5) **Ans: 5**
121. A typist made 2.8 mistakes on an average per page. Find the probability that in the page typed by her i) There is no mistakes ii) There are two or less mistakes
122. A car hire firm has 2 cars that it hires out day by day. The number of demands for a car on each day is distributed as Poisson distribution with mean 1.5. Calculate the proportion of days on which i) Neither car is used ii) some demand is refused
123. At a checkout counter customers arrive at an average number of 1.5 per minute. Find the probability that:
    a. at most four will arrive in any given minute
    b. At least 3 will arrive during an interval 2 minutes.
124. The cheques received by a bank follows Poisson distribution with an average 6 bad cheques per day, what are the probabilities that it will receive
    a. 4 bad cheques per day
    b. 10 bad cheques over any 2 consecutive days
125. If a mean number of serious accidents per month in a large factory are one find the probability that in the current year there will be at least 10 accidents. The annual serious accidents in the factory follow a Poisson distribution
126. An office switch board receives an average of 4 emergency calls in a 10 minute interval. What is the probability that
    a. there is no call in a 10 minutes interval
    b. Thereare at most 2 calls in 5 minutes interval?
127. A telephone exchange receives on an average 4 calls per minutes. Find the probability on the basis of Poisson distribution of**PU2005 Ans:0.0183, 0.238, 0.628, 0.3713**
    a. No call at all
    b. 2 or less cal per minute
    c. Up to 4 calls per minute
    d. More than 4 calls per minute
128. Customers arrive at complaint department of a store at the rate of 5 per hour. If arrivals follow a Poisson distribution. Calculate the probability that**PU2013 Ans:0.00674,0.959,0.125**
    a. No customer will arrive in an hour
    b. two or three customers will arrive in an hour
    c. At least three customers will arrive in an hour **Ans: 0.0497, 0.448, 0.57759 PU2017**
129. Customers arrive at complaint department of a store at the rate of 3 per hour. If arrivals follow a Poisson distribution. Calculate the probability that **PU2013 Ans:0.00674,0.959,0.125**
    a. No customer will arrive in an hour
    b. At least two customers will arrive in an hour
130. Customers arrive at the counter of ABC bank at the rate of 4 per hour. If arrivals follow a Poisson distribution. Calculate the probability that
    a. At least 3 customers will arrive in an hour
    b. No customer that will arrive in a 30 minute period
131. The number of accident in a year attributed to taxi drivers in a city is Poisson distribution with mean 3.Out of 1000 taxi drivers, find approximately the number of drivers with
    a. More than 3 accidents in a year
    b. less than 2 accidents
132. Messages arrive at an electronic message center at random times, with an average of 9 messages per hour.

I) What is the probability of receiving at least five messages during the next hour?
II) What is the probability of receiving exactly five messages during the next hour?**PU2018**

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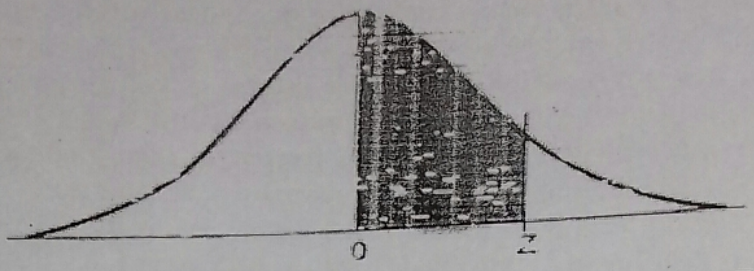

This table presents the area between the mean and the Z score . When Z=1.96, the shaded area is 0.4750.

Areas Under the Standard Normal Curve

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4999	.4999	.4999	4999	.4999	.4999	.4999	.4999
3.9	.5000									

Source: Adapted by permission from *Statistical Methods* by George W. Snedecor and William G. Cochran, sixth editi
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Normal Probability Distribution

133. The r.v. $X \sim N(300, 25)$ find i) $P(x > 305)$ ii) $P(x < 291)$ iii) $P(x < 312)$ iv) $P(x > 286)$

134. Given a Normal distribution with mean 100 and sd 10. What is the probability that
   a. $X > 75$
   b. $X < 70$
   c. $X > 112$ **Ans:0.9938, 0.00135, 0.1151, 0.0606, 0.1815**
   d. $75 < X < 85$
   e. $X < 80$ or $X > 110$

135. The owner of a fish market determined that the average weight for a catfish is 3.2 pounds with a standard deviation of 0.8 pounds. Assuming the weights of catfish is normally distributed.**Ans:0.5865, 2.184**
   a. What is the probability that a randomly selected catfish will weigh between 3 and 5 pounds?
   b. Above what weight (in pounds) do 89.80% of the weights occur?

136. The average diameter of a sample of 1000 pipes is 2.6 inches and standard deviation is 0.55. Assuming that the diameter of pipes is normally distributed. Find the number of pipes of diameter **PU2018**
   I) Greater than 2 inches.
   II) Between 2 and 3.1 inches

137. The time taken by a delivery man to deliver milk is normally distributed with mean 12 minutes and standard deviation 2 minutes. He delivers milk every day. Estimate the number of days in a year (take 1 year = 365 days) when he takes
   a. longer than 17 minutes
   b. between 9 and 13 minutes

138. Diameters of bolts produced by a particular machine are normally distributed with mean 0.76 cm and standard deviation 0.012 cm. specifications call for diameters from 0.72 cm to 0.78 cm **PU2018**
   a) What percentage of bolts will meet these specifications?
   b) What percentage of bolts will be smaller than 0.73 cm?

139. As a result of tests on 20000 electric bulbs manufactured by a company it was found that the life time of the bulb was normally distributed with an average of 2040 hours and the standard deviation of 60 hours. On the basis of the information, estimate the number of bulbs that are expected to burn for less than 1960 hours **Ans:1836**

140. A statistical analysis of 100 long distance telephone calls made from the headquarters of the bricks an clicks. Computer Corporation indicates that the length of these calls is normally distributed with mean seconds and standard deviation 40 seconds.**Ans:6.68%, 0.8664, 0.1336, 146.8**
   a. What percentage of these calls lasted less than 180 seconds?
   b. What is the probability that a particular call lasted between 180 and 300 seconds?
   c. How many calls lasted less than 180 seconds or more than 300 seconds?

141. Incomes of a group of 10,000 persons were found to be normallydistributed with mean of Rs. 5200 and standard deviation Rs. 600. Find **Ans:6687, 6163**
   a. The number of persons having income between Rs. 4000 and Rs. 5500?
   b. The lowest income of the richest 600?

142. In a certain examination test 2000 students appeared in Statistics. The average marks obtained were 50% and the standard deviation was 5%. How many students do you expect to obtain more than 60% marks? What are the minimum marks of the top 100 students? Assume that the marks are normally distributed. **PU2016 Ans: 46, 58.2%**

143. A statistical analysis of 1000 long distance telephone calls made from the headquarters of the Bricks and Clicks Computer Corporation indicates that the length of these calls is normally distributed with mean 240 seconds and standard deviation 40 seconds.
   a. What percentages of these calls lasted less than 180 seconds?
   b. What is the length of a particular call if only 1% of all calls are shorter?

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144. The manager of a small postal substation is trying to qualify variation in the weekly demand for mailing bags. She has decided to assume that this demand is normally distributed. She knows that an average 100 bags are purchased weekly and that 90% of the time weekly demand is below 11

a) What is the standard deviation of this distribution?**Ans:11.63, 119.18**

b) The manager wants to stock enough bags each week so that the probability of running out a quiet of bags is no higher than 0.05. What is the lowest such stock value?

145. A statistical analysis of 1000 long distance telephone calls made from the headquarters of the Bricks and Clicks Computer Corporation indicates that the length of these calls is normally distributed with mean 240 seconds and standard deviation 40 seconds.

a) What percentages of these calls lasted less than 180 seconds?

b) What is the length of a particular call if only 1% of all calls are shorter?

146. Toby's Trucking Company determined that on an annual basis the distance travelled per truck is normally distributed with a mean of 50 thousand miles and standard deviation of 12 thousand mil

a. What proportion of trucks can be expected to travel between 34 and 50 thousand miles in year?

b. What percentage of trucks can be expected to travel either below 30 or above 60 thousand miles in the year?**Ans: 40.82%, 25.08%**

147. In a distribution exactly normal, 31% of the items are under 45 and 8% of the items are over What are the mean and standard deviation of the distribution?**Ans:50,10**

148. In final examination of statistics subject, the marks obtained by students were observed to normally distributed with mean 73 and standard deviation of 8.**Ans:0.9484, 0.9318, 86.2**

a) What is the probability of students getting marks more than 60?

b) What percentage of students scored marks between 60 to 90?

c) Only 5% of the students appearing in the exam score higher than which marks?

149. The life time in hours of a certain electrical equipment has the normal distribution with mean hours and standard deviation 16 hours**Pu2012 Ans:0.1056 0.883 0.3579**

a. What is the probability that equipment lasts at least 100 hours?

b. What is the probability that equipment lasts at most 99 hours?

c. What is the probability that equipment lasts between 73 hours and 88 hours?

150. In a test administered to500 students, the average score was 42 and standard deviation 24. Find

a. The number of students exceeding a score 50.**Ans:185,177,62.16**

b. The number of students lying between 32 and 54.

c. The value of score exceeded by the top 100 student.

151. Assuming the height of soldier is normally distributed with mean 68.22 inches and variance sq. inches. How many soldiers in a regiment of 1000 would you expect to be over six tall?**Ans:125PU2009**

152. The lengths of certain items follow a normal distribution with mean μ cm and standard deviati cm. It is known that 4.78% of the items have a length greater than 82 cm. Find the value of the μ:

153. Last year, the BBA students of a campus scored marks in Business statistics, which are distrib normally with mean 72 and standard deviation 6. This year 300 students are appearing in this pap

a. Estimate the number of students who scores more than 80 marks.

b. If 65 is the pass mark, estimate the number of students who fails.

154. The distribution of the wages of the employees in a firm was perfectly normal and the number o workers employed in the organization was 4,000.the mean and standard deviation was calculated Rs. 800 and Rs. 200 respectively. On the basis of the information estimate **PU2005Ans:1532, 38.3%, 15.87%**

a. The number of workers getting salary between Rs. 700 and Rs. 900.

b. % of workers getting salary between Rs. 700 and 900.

c. % of workers getting salary above Rs. 1000.

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155. Unisys.com is one of the most frequented business to business websites; assume that the length of a visit on the Unisys websites is distributed as a normal random variable with a mean of 65.7 minutes and a standard deviation of 15 minutes.PU2015 **Ans:0.0526, 52.95**
   a. What is the probability that a randomly selected visit will last more than 90 minutes?
   b. Only 20% of the visits will last less than how many minutes?

156. In an examination 15% of the candidates got first class (60% marks or above), while 40% failed (securing below 40% marks). Assuming the marks to be normally distributed, estimate the mean and standard deviation. **PU2016 Ans:44, 15.38**

157. In an examination 15% of the student's marks more than 60 and 40% of the students got marks less than 40. Assuming the marks to be normally distributed, find the mean and standard deviation. **PU2019 Ans:44, 15.38**

158. In an examination, 30% of the candidates obtained marks below 40 and 10% of the candidates got above 75 marks. Assuming that the marks are normally distributed, find mean and standard deviation of the distribution. **PU2014 Ans:50 19**

159. Of a large group of men, 5% are less than 60 inches in height and 10% are above 72 inches in height. If height of men is normally distributed, find the mean height and standard deviation. **PU2013 Ans: 66.74, 4.07**

160. Batteries last an average of 60 days with standard deviation of 10 days. Sixty batteries are bought. What is the probability that**PU2015**
   a. Mean life of all 60 batteries exceed 61 days
   b. Mean life differs from 60 days by more than 3 days.

161. The income of a group of 20,000 persons was found to be normally distributed with mean Rs. 7,500 and standard deviation Rs. 500. Find **Ans:7925 6875**
   a. The lowest income of richest 20% of persons
   b. The highest income of poorest 10% of persons.

162. A sample of 600 mobile of a certain Brand is tested to find the length of Battery. After each full charge, produced result of average 12 hours and standard deviation of 4.5 hours. Assuming the data to be Normally distributed, find the number of mobiles that are expected to have life after each full charge:
   a) More than 15 hours
   b) Between 20 and 14 hours
   c) Top 10% performer (on the bases of battery life) **PU2017 Ans:151,176**

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Uniform, Gamma and Log Normal distribution

163. Busses arrive at a specified stop at 15 minute intervals starting at 7 am. That is, they arrive at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits.PU2015 Ans:1/3 1/5
    a. Less than 5 minutes for a bus
    b. At least 12 minutes for a bus
164. Suppose the time it takes a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 minutes. PU2014 Ans:1.85 0.041 0.714 1.4286(
    1.5)
    a. What is the mean and variance of the time it takes an operator to fill out the form?
    b. What is the probability that it will take less than two minutes to fill out the form?
    c. Determine the cumulative distribution function of the time it takes to fill out the form.
165. The time (in hours) required to repair a machine is exponentially distributed with parameter $\lambda=\frac{1}{3}$. What is the probability that the repair time exceeds 3 hours? Also find the probability that machine will required within 3 hours. PU2015 0.367, 0.632
166. Suppose that during rainy season on a tropical island the length of the shower has an exponential distribution, with parameter $\lambda=2$, time being measure in minutes. What is the probability that a shower will last more than three minutes?Ans: 0.00247
167. Suppose that the service life of a semiconductor is a random variable having exponential distribution with $\lambda=\frac{1}{50}$ hours. Find the probability that such a semiconductor will still be operating condition after 100 hours. Ans:01353 PU2017
168. The life time of semiconductor laser has a log normal distribution with $\mu=10$ hours and $\sigma=1.5$ hoursPU2016
    a. What is the probability that the life time exceeds 10,000 hours?
    b. What life time is exceeded by 99% of laser?Ans:668 0.7019
169. The length of time (in seconds) that a user views a page on website before moving to another page is log normal random variable with parameters $\mu=0.5$ and $\sigma^2=1$ PU2016
    a. What is the probability that a page is viewed for more than 10 seconds?
    b. What is the length of time 50% of user view the page?
    c. What is the mean and standard deviation of the time until a user moves from the page?
170. In a certain city, the daily consumption of electric power (in millions of kilowatt hours) can be treated as a random variable having a gamma distribution with $\alpha=3$ and $\beta=2$. If the power plant of this city has a daily capacity of 12 million kilowatt hours. What is the mean of this gamma distribution? What is the probability that this power supply will be inadequate on any given day?PU2014 Ans:0.0619
171. The magnitude of earthquakes recorded in Sindhupalchowk of Nepal during the year 2072 can be modeled as having an exponential distribution with mean 4.5, as measured on the Richter scale. Suppose that the magnitude of earthquakes striking the region has a gamma distribution with $\alpha=$ and $\beta=3$
    a. What is the mean magnitude of earthquakes striking the region?
    b. What is the probability that the magnitude an earthquake striking the region will exceed the Richter scale?
    c. What is the probability that an earthquake striking the regions will fall between 5 and 6 on the Richter scale?PU2016 Ans:4.5 0.3292
172. In a certain city the daily consumption of water (in millions of gallons) follow approximately a gamma distribution with parameter $\alpha=2$ and $\beta=3$. If the daily capacity of this city is 9 million gal of water, what is the probability that on any given day the water supply is inadequate?
    Ans:0.19914
173. Suppose that the life time of a certain kind of computer is random variable X having the Gamma distribution with α and $\beta=6$. Find Ans: 12, 0.5036
    a. Mean lifetime of computers
    b. The probability that such a computer will last more than 10 years?

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Joint Distribution

174. Suppose X and Y have the following joint probability mass function

Y \ X	2	3
1	0.1	0.15
2	0.2	0.3
3	0.1	0.15

Find **Ans: 0.4 0.6 0.24 0.5 2/5, 3/5, ¼, ½, ¼, 0.3, 0.75**

a. The marginal distribution of X and Y
b. The conditional probability of X given that Y=2
c. The conditional probability of Y given that X=3
d. F(2,2) and F(3,2)
e. Are X and Y independent?
f. Variance of X and variance of Y

175. For the joint probability distribution of two random variable X and Y given below:

X \ Y	1	2	3
1	5/27	4/27	2/27
2	1/27	3/27	3/27
3	3/27	4/27	2/27

a. Find the marginal distribution function of X and Y
b. Find the mean and variance of X and Y

176. For the joint probability distribution function of random variable X and Y given below

X \ Y	1	2	3	4
1	4/36	3/36	2/36	1/36
2	1/36	3/36	3/36	2/36
3	5/36	1/36	1/36	1/36
4	1/36	2/36	1/36	5/36

a. Find the marginal distribution of X and Y
b. Find E(X) and E(Y)

177. Suppose the joint probability function of random variables X and Y is given by **PU2013**

$$f(x,y) = 4xy, \quad 0 < x < 1, 0 < y < 1$$
$$= 0, \quad \text{otherwise}$$

a. Verify that the joint probability function is a joint pdf
b. Find P(X<0.5 and Y<0.8) **Ans: 0.16**
c. Find the marginal distribution of X and Y **Ans: 2x, 2y**
d. Find the conditional distribution of X and Y **Ans: 2x and 2y**
e. Is X and Y are independent? **Ans: independent**
f. Find E(X), E(Y), V(X) and V(Y) **Ans: 2/3, 2/3, 1/18, 1/18**

178. The joint probability density function of a two dimensional random variable (X,Y) is given by **PU2015**

$$f(x,y) = \frac{1}{8}(6 - x - y), \quad 0 < x < 2, 2 < y < 4$$
$$0, \quad \text{otherwise}$$

Find

a. P(X<1∩Y<3) Ans: 3/8, 5/24, 3/5, $\frac{3-x}{4}, \frac{5-y}{4}$
b. P(X+Y<3)
c. P(X<1/Y<3)
d. Find the marginal pdf of X and Y

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b.P(X≥1,Y≥1)

c. Are X and Y dependent or independent?

187. If the joint probability density of two random variables is given by PU2005

$$f(x_1, x_2) = 6e^{-2x_1 - 3x_2} \text{ for } x_1 \geq 0, x_2 \geq 0$$

Find

a.Marginal density function of X and Y

b.Are X_1 and X_2 independent?

188. Two random variables X and Y have the following probability density function. PU2017

$$f(x,y) = k(6 - x - y) \quad 0 < x < 2, \quad 0 < y < 2$$
$$= 0 \qquad \text{otherwise}$$

Find

a. The constant k
b. Marginal density function of X and Y
c. E(X) and V(X)
d. Are X and Y independent?

189. A two dimensional r.v. (X,Y) have following probability density function PU2019

$$f(x,y) = \frac{2}{3}(x + y) \qquad 0 < x < 1, \qquad 0 < y < 1$$

a) Find marginal pdf of X and Y
b) Find conditional pdf of X given Y
c) Check whether X and Y are independent or not.

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179. Two random variables X and Y have the following probability density function.PU2009

$$f(x,y) = k(4-x-y) \quad 0<x<2,\ 0<y<2$$
$$= 0 \qquad \text{otherwise}$$

Find

a. The constant k Ans: 1/8, $\frac{3-x}{4}, \frac{3-y}{4}, \frac{4-x-y}{2(3-y)}, \frac{5}{6}, \frac{11}{26}$,
b. Marginal density function of X and Y
c. Conditional probability density function of X given Y
d. E(X) and V(X)

180. Two dimensional random variables (X,Y) have the joint density

$$f(x,y) = 8xy, \quad 0<x<1, \qquad x<y<1$$

a. Find marginal pdf of random variables X and Y Ans: $4x(1-x^2), 4y^3, \frac{2x}{y^2}, \frac{2y}{1-x^2}$
b. Find conditional pdf of random variables X given Y
c. Find conditional pdf of random variables X and Y

181. The joint probability density function of two continuous random variablesPU2016

$$f(x,y) = Axy \qquad 0<x<1 \qquad 0<y<x$$

a. Find A Ans: 8, $4x^3$, $4y(1-y^2)$, $\frac{2y}{x^2}, \frac{2x}{1-y^2}$, *are not independent*
b. Find marginal density function of X and Y
c. Find conditional density function of Y given X=x and conditional density function of X given Y=y
d. Check for independence of X and Y

182. The joint pdf of X and Y is

$$f(x,y) = kxy \qquad 1<x<y<2$$

a. Find the constant k Ans: 8/9, $\frac{4}{9}x(4-x^2), \frac{4}{9}y(y^2-1)$
b. Find marginal pdf of X and Y

183. If two random variables have the joint probability density function

$$f(x,y) = \frac{2}{3}(x+2y) \qquad 0<x<1, \qquad 0<y<1$$

a. Find marginal density function of X and Y
b. Find the conditional density of X given that Y=y
c. Are random variables independent

Ans: $\frac{2}{3}(x+1), \frac{2}{3}\left(\frac{1}{2}+2y\right)$, *are not independent*PU2017

184. Joint distribution of X and Y are given byPU2014

$$f(x,y) = 4xye^{-(x^2+y^2)}, x>0 \text{ and } y>0$$

a. Test whether X and Y are independent? Ans : independent, $2xe^{-x^2}$
b. Find the conditional density of X given Y=y

185. Two random variable X and Y have the following pdfPU2003

$$f(x,y) = 2-x-y \quad 0 \le x \le 1, 0 \le y \le 1$$
$$0 \qquad \text{otherwise}$$

Find

a. Marginal probability density function of X and Y
b. V(X) and V(Y)
c. Covariance between X and Y

186. The joint probability density function of a two dimensional random variable (X,Y) is given by PU2014

$$f(x,y) = \frac{1}{8}(4-x-y), \quad 0<x<2, 0<y<2$$
$$0 \qquad , \qquad \text{otherwise}$$

Find

a. The marginal distribution of X and Y

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Table T Critical Values of the t Distribution

df	One-Tail = .4 Two-Tail = .8	.25 .5	.1 .2	.05 .1	.025 .05	.01 .02	.005 .01	.0025 .005	.001 .002	.0005 .001
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.6
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	14.089	22.327	31.5
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	7.453	10.214	12.9
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.6
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.8
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.9
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.4
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.0
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.7
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.5
11	0.260	0.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.4
12	0.259	0.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.3
13	0.259	0.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.2
14	0.258	0.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.1
15	0.258	0.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.0
16	0.258	0.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.0
17	0.257	0.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.9
18	0.257	0.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.9
19	0.257	0.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.9
20	0.257	0.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.8
21	0.257	0.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.8
22	0.256	0.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.8
23	0.256	0.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.8
24	0.256	0.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.7
25	0.256	0.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.7
26	0.256	0.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.7
27	0.256	0.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.7
28	0.256	0.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.7
29	0.256	0.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.6
30	0.256	0.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.6
40	0.255	0.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.5
60	0.254	0.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.4
120	0.254	0.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.3
∞	0.253	0.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.2

Source: From *Biometrika Tables for Statisticians*, Vol. 1, Third Edition, edited by E. S. Pearson and H. O. Hartley, 1966, p. 146. Reprinted by permission of the Biometrika Trustees.

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.001 .002	.0005 .001
318.31	636.62
22.327	31.59
10.214	12.92
7.173	8.61
5.893	6.86
5.208	5.95
4.785	5.40
4.501	5.04
4.297	4.78
4.144	4.58
4.025	4.42
3.930	4.31
3.852	4.22
3.787	4.14
3.733	4.07
3.686	4.01
3.646	3.96
3.610	3.92
3.579	3.88
3.552	3.85
3.527	3.82
3.505	3.79
3.485	3.77
3.467	3.74
3.450	3.72
3.435	3.70
3.421	3.68
3.408	3.66
3.396	3.64
3.385	3.63
3.307	3.54
3.232	3.46
3.160	3.38
3.090	3.29

3.1966), p. 146.

Estimation Theory

190. A population consists of the value 7, 6, 8, 4, 10. Prove that sample mean of size 3 is an unbiased estimator of the population mean. **PU2015**

191. A population consists of five members 2, 3, 6, 8 and 11.
   a) Draw all possible sample of size 2 that can be drawn from the population without replacement.
   b) Find mean and variance of population
   c) Find mean of sampling distribution of means and show that it is equal to the population mean.**PU2019**

192. A population consists of the value 3, 6, 8, 9, 12 and 15. Prove that sample mean of size 3 is an unbiased estimator of the population mean. **PU2015**

193. A population consisits of the values 2, 4. 5,6,7,8 and 9. Prove that sample mean of size 3 is an unbiased estimator of the population mean. **PU2012**

194. From a population of 540, a sample of 60 individuals is taken. From this sample, the mean is found to be 6.2 and standard deviation 1.368. **PU2015 Ans: 0.167, (5.86,6.54)**
   a. Find the estimated standard error of the mean
   b. Construct 96% confidence interval for the mean

195. A sample of 30 households drawn from a city area containing 14845 total households. It was known that the average household's size is 3.4 and standard deviation is 1.22. Estimate the total city population at 95% confidence level.**Ans:(43999, 56947)**

196. A sample of 500 bulbs of a company showed an average life of 1400 hours with standard deviation of 30 hours. Obtain a. 95% and b. 99% confidence limits for population mean. **Ans:(1397.37, 1402.63)**

197. A manufacturer of computer paper has a production process that operates continuously throughout an entire production shift. The paper is expected to have an average length of 11 inches and the standard deviation is known to be 0.01 inch. Suppose a random sample of 100 sheets and the average paper length is found to be 10.998 inches. Set up 95% and 99% confidence interval estimate of the population average paper length.**Ans: (10.996, 11)**

198. A sample of 400 students taking entrance examination of MBBS revealed an average score of 56 and sample standard deviation of 10. Construct a 95% and 99% confidence interval for the population mean.
Ans@55.02,56.98) (54.71,57.29)

199. A sample of 500 students taking entrance examination of MBBS revealed an average score of 56 and sample standard deviation of 10. Construct a 95% and 90% confidence interval for the population mean.

200. The manager of a paint supply store wants to estimate the actual amount of paint contained in one gallon cans purchased from a nationally known manufacturer. It is known from the manufacturer's specifications that the standard deviation of the amount of paint is equal to 0.05 gallon. A random sample of 64 cans is selected and the mean amount of paint per one gallon can is 0.995 gallon. Construct 95% confidence interval estimation of the population mean amount of paint included in a one gallon can.**Ans: (0.9872, 1.0028)**

201. The quality control manager at a light bulb factory needs to estimate the mean life of a large shipment of life bulbs. The process SD is known to be 100 hours. A random sample of 64 light bulbs indicated a sample mean life of 350 hours.**Ans:12.5,(325.5,374.5) PU2018**
   a) Obtain the standard error of mean and interpret its meaning
   b) Set up 95% confidence interval of the true population mean life of bulbs in this shipment.
   c) Do you think that manufacturer has the right to state the light bulbs last an average of 390 hours? **PU2017**

202. The Kathmandu High School nurse is interested in knowing the average height of seniors at this school, but he does not have enough time to examine the records of all 430 seniors. She randomly selects 48 students. She finds the sample mean to be 64.5 inches and the standard deviation to be 2.3 inches. **Ans:0.313, (63.98,65.016)**
   a. Find the estimated standard error of the mean.

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b. Construct a 90% confidence interval for the mean.

203. A certain University in USA is contemplating switching from the semester system to an annu system. The administration conducts a survey of a random sample of 400 management students a finds that 240 of them prefer to remain on the semester system. Construct a 95% confidence interv for the true proportion of all management students who would prefer to remain on the semest system Ans: (0.552, 0.648)

204. A manufacturer of electronic calculators takes a random sample of 1200 calculators and finds that there are eight defective units. Construct a 95% confidence interval on the population proportion **PU2014**

205. In a survey of 763 women who had started their own business, 229 said that they launched their business for greater freedom. Only 99 indicated that a desire to make more money drove them to start their business.

a. Set up 90% confidence interval estimate of the population proportion of women who start new business to gain freedom.**Ans(0.272 0.327)**

b. Set up 90% confidence interval estimate of the population proportion of women who st new business to earn more money.**Ans(0.1097 0.1499)**

206. A random sample of 700 units from a large consignment showed that 200 were damaged. Find a 95% and b. 98% confidence limits for the proportion of damaged units in the consignment.

207. A random sample of 500 apples was taken from a large consignment and 60 were found to be damage. Construct 95% confidence interval for the good apples in the population.

208. A random sample of 100 items taken from a large batch of articles contains 85 defective items. Determine the 95% and 98% confidence limits for the proportion of defective items in the batch.

209. General Cinema sampled 55 people who viewed Ghost hunter and asked them whether they planned to see it again. Only 10 of them believed the film was worthy of a second look. **PU2009**

a. Estimate the standard error of the proportion of moviegoers who will view the film a secon time.

b. Construct 95% confidence interval for this proportion. **Ans:0.0539 (0.0944 0.3056)**

210. A random sample of 500 apples was taken from a large consignment and 65 of them were found t be bad. Assign the limits within which the percentage of good apples in the consignment will lie a 95% to confidence level.**Ans: (0.8406, 0.8994)**

211. The probability of getting bad apple in sampling with sample size 400 from a large consignment 0.12. Obtain the 98% confidence limit for the bad apple in the consignment. **Ans: (0.08214, 0.15785)**

212. In a random sample of 300 households in a city 223 have computer. Set an approximate 95% and 99% limits to the true value of proportion of households with computer in the whole city. **PU2018**

213. A random sample of 100 patients suffering from a certain disease was given serum treatment. It w observed that 75% were released of the disease. Find 95% confidence limit for the percentage of patient cured. **PU2005**

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214. A certain Cinema hall sampled 55 people who viewed "DHOOM 3" and asked them whether they planned to see it again. Only 10 of them believed the film was worthy of a second look. **PU2009**

Ans: 0.052, (0.0963, 0.2673)

a. Estimate the standard error of the proportion of moviegoers who will view the film a secon time.

b. Construct 90% confidence interval for this proportion

215. A sample of 100 engineers from small construction companies and a sample of 200 engineers fro large construction companies were interviewed about having expense controls in place for use of cellular phones. Of the small companies, 47 had controls in place whereas 79 of the large compani had controls. Determine the 95% confidence in population proportion in expense controls in place for large and small company's engineers. Ans:

216. Hotel's manager in Kathmandu wants to know the hotels average daily registration. The followin table presents the numbers of guest registered each of 27 randomly selected days. Calculate the

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sample mean, standard errors of mean and 95% confidence limits of population mean. PU2016
Ans(56.12,59.36)

61	57	53	60	64	57	54	58	63
61	50	59	50	60	57	58	62	63
60	54	54	61	51	53	62	57	60

217. A random sample of 12 values from a normal population showed a mean of 31.9 inches and sum of square of deviation from this mean equal to 125 square inches. Obtain 95% confidence interval for μ. PU2012 Ans(29.75,34.04)

218. A random sample of 7 envelopes is taken from letter box of a post office and their weights in grams are found to be 12.1, 11.9, 12.3, 11.9, 12.1, 12.4 and 12.1. Find 99% confidence limits for the mean weight of envelops received at that post office. Ans: (11.9025, 12.3974)

219. A random sample of 8 envelopes is taken from letter box of a post office and their weights in grams are found to be 12.1, 11.9, 12.3, 11.9, 12.1, 12.4, 11.5 and 12.1. Find 99% fudicial limits for the mean weight of envelops received at that post office. Ans: (11.69, 12.38)

220. A sample of 9 shafts is inspected from a population line. The following measurements are the diameters in (mm) of shafts: 45.01, 45.02, 45.021, 45.015, 45.019, 45.018, 45.02, 45.023, and 45.005. Estimate the 95% confidence interval within which the true diameter of the shaft line lie. Ans:(45.01328, 45.02333)

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Hypothesis Testing

221. A random sample of 900 items is drawn from a normal population whose mean is 3.25 cm and whose standard deviation is 2.61 cm. if the sample mean is 3.5 cm. can the sample be regarded as true random sample at 5% level of significance? Also construct 95% confidence interval for the true mean in the population **PU2003** ***Ans:2.8726reject***

222. An educator claims that the average I.Q. of college students is at most 110, and that the study mean to test this claim 150 college students selected at random had an average I.Q. of 111.2 with a standard deviation of 7.2. Use a level of significance of 0.01 to test the claim of educator.
Ans: 2.04accept

223. You are the manager of a restaurant that delivers pizza to the college dormitory rooms. You have just changed to your delivery process in an effort to reduce the mean time between the order and completion of delivery current 25 minutes. From past experience, you can assume that the population standard deviation is 6 minutes. A sample of 36 orders using the new delivery process yields a sample mean of 22.4 minutes. At the 5% level of significance, is there evidence that the mean delivery time has been reduced below the population mean value of 25 minutes?**Ans: -2.6**

224. A sample of 100 tires is taken from a lot. The mean life of tires is found to be 39350 km with standard deviation of 3260 km. could the sample come from a population with mean life of 40,000 km at 5% level of significance?

225. A sample of 100 students is taken from a large population. The mean height of those students is inches and standard deviation of 4 inches. Can it reasonably be regarded that in the population mean height is less than 66 inches at 5% level of significance? **Ans: -5**

226. The strength of steel wire made by an existing process is normally distributed with mean of and standard deviation of 150. A batch of wire is made by a new process, and a random sample consisting of 25 measurements gives an average strength of 1312. Assume that the standard deviation does not change. Is there evidence at 1% level of significance that the new process gives larger mean strength than the old? **PU2019**

227. A machine is used to cut wheels of cheese into blocks of specified weight. On the basis of experience it has been observed that the weight of the blocks is normally distributed with a standard deviation of 0.3 kg. The machine is currently set to cut the block of weight 12 kg. a sample blocks if found to have an average block of weight 12.25 kg. Set up the null hypothesis determine whether it is a one tailed or two tailed test. Use the critical value approach to determine whether the cutting machine needs to be recalibrated? **PU2015*Ans: 2tail, reject Ho***

228. A sample of 900 members has a mean 3.4 cms and sd of 2.61 cms. Can the sample be regarded as one drawn from a population with mean 3.25 cms?

229. The Glen Valley steel company manufactures steel bars. If the production process is working properly, it turns out steel bars with mean length of at least 2.8 feet with a standard deviation of 0.3 feet. Longer steel bars can be used or altered, but shorter bars must be scraped. A sample of 25 bars is selected from the production line. The sample indicates a mean length of 2.73 feet. The company wants to determine whether the production equipment needs to be adjusted. Test at 5% level of significance. **Ans:-1.75**

230. A researcher wishes to test the claim that the average age of lifeguards in Ocean city is greater than 24 years. She selects a sample of 36 guards and finds the mean of the sample to be 24.7 years, with a standard deviation of 2 years. Is there evidence to support the claim at 0.05?

231. A study on expenditure behavior of tourists in Nepal revealed that the average expenditure of 100 European tourists per day is NPs. 9200 with standard deviation of NRs 600. Also, the average expenditure of 100 Americans tourists per day is NRs. 10,000 with standard deviation of NRs.500. Can you claim that the expenditure behavior of European and American tourists are same?**Ans:-10.2**

232. Cortical level determinations were made on two samples of women at child birth. Group first subjects underwent emergency cesarean section following induced labor. Group second subjects delivers by either cesarean section or the vaginal route following spontaneous labor. The sample sizes, mean cortical levels and standard were as follows:

Sample	Size of sample	means	Standard deviations

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Group First	110	435	65
Group Second	120	645	80 **Ans:21.9**

Do these data provide sufficient evidence to indicate a difference in the means cortical levels in the populations represented?

233. A company claims that its light bulbs are superior to those its main competitor. If a study showed that a sample of $n_1=40$ of its bulbs had a mean life time of 647 hours of continuous use with a standard deviation of 27 hours, while a sample of $n_2=40$ bulbs made by its main competitors had a mean life time of 638 hours of continuous use with a standard deviation of 31 hours, does this substantiate the claim at the 0.05 level of significance? **PU2012*Ans: 1.38***

234. A potential buyer of light bulbs bought 50 bulbs of each of two brands upon testing these bulbs, he found that Brand A has a mean life of 1282 hours with a standard deviation of 80 hours whereas B had a mean life of 1208 hours with a standard deviation of 94 hours can the buyers be quite certain that the two brands differ in quality? **PU2005 Ans:4.23**

235. The average height of 50 students who showed athletic interest was 68.2 inches with standard deviation of 2.5 inches, while another set of 50 students who showed no athletic interest had the average height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that athletics interest makes a student taller. **PU2009 Ans:1.318**

236. In a certain factory there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 grams with standard deviation of 12 grams, while the corresponding figures in a sample of 400 itemsfrom the other process are 124 and 14. Test whether two mean weights differs significantly or not at 10% level of significance.**PU2015 Ans:3.874**

237. The average weekly wages of a sample of 200 workers in industry A was Rs. 1150 with standard deviation of Rs. 100. The weekly wages of a sample of 300 workers in industry B was Rs. 1,000 with standard deviation of Rs. 50. Can we consider the weekly wages paid by industry A is higher than those paid by industry B? **PU2013 Ans:19.64**

238. An insurance company, based on past experience, estimates the mean damage for a natural disaster in its area is $5,000. After introducing several plans to prevent loss, it randomly samples 200 policyholders and finds the mean amount was $4,800 with a standard deviation of $1,300. Does it appear the prevention plans were effective in reducing the mean amount of claim? Use 0.05 significance level.

239. The average monthly salaries of two groups of workers have to be compared. A random sample of 36 workers from group A has a monthly average of Rs 6434 with standard deviation Rs 19 and that of 64 workers from group B is Rs 6408 with standard deviation of Rs 17. Do the data present sufficient evidence to indicate a difference in the average salary for the two group of workers at 1% level of significance?**Ans:6.82**

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240. The average hourly wage of a sample of 150 workers in City A was Rs. 25.6 with a standard deviation of 10.8. The average hourly wage of a sample of 200 workers in City B was Rs. 28.7 with a standard deviation of Rs. 12.8. Can an applicant safely assume that the hourly wage paid in City B is higher than those paid in city A? **Ans:1.035**

241. An examination was given to 50 students at college A and 60 students at college B. at A mean grade was 75 with sd of 9. At B, mean grade was 79 with sd 7. Is there significance difference between the performance of students A and those at B given $\alpha=0.05$ **Ans:2.56**

242. In a certain factory, there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 grams with a standard deviation of 12 grams, while the corresponding figures in a sample of 400 items from the other process are 124 and 14. Find the standard error of the difference of means and also test whether the two mean weights differ significantly or not at 10% level of significance.

243. In a sample of 1000 people in Kathmandu district, 540 speak Nepali and rest speaks Newari. Can we assume that both languages are equally popular in this district at 1% and 5% level of significance? Ans:2.53

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244. The audit is conducting on a sample of 75 of the reimbursements of which 12 visits were incorr amount of reimbursement was provided. At 0.05 level of significance is there evidence that the proportion of incorrect reimbursements in the population proportion is greater than 0.10? Ans:1.7
245. A manufacture company claims that at least 95% of his production confirms to be good. Examination of sample of 1000 pieces of produced goods 60 was defective. Test his claim at a significance level of 1% Ans: -1.4588
246. Consider the case of pharmaceutical Manufacturing Company testing two new compounds inten to reduce blood pressure level. The compounds are administered two different sets of laboratory animals. In group A 71 out of 100 animals tested response to drug first with lower blood pressure levels. In group B, 58 out of 90 animals tested respond to drug second with lower blood pressure level. The company wants to test at 0.05 levels whether there is a difference between the effectiveness of these two drugs **Ans:0.96**
247. It is believed that the proportion of dissatisfied government employees is lower in Kathmandu valley than those in outside valley. In a sample of 500 government employees of Kathmandu valley 145 were found dissatisfied. In another sample of 300 government employees from outside valley 105 were found dissatisfied. Test the belief at 0.01 level of significance **Ans: -1.77**
248. A random sample of 200 bolts manufactured by machine A and of 100 bolts manufactured by machine B showed 19 and 5 defective bolts respectively. Test the hypothesis that machine B is performing better than A at 5% level of significance **PU2018**
249. 1000 articles from factory A are examined and found to contain 3% defectives 1500 similar artic from factory B are found to contain only 2% defectives. Would you conclude that the products of factory B are superior to the factory A? **PU2003**
250. A study shows that 16 of 200 tractors produced on one assembly line required extensive adjustm before likely they could be shipped, while same was true for 14 of 400 tractors produced on anoth assembly line. At 0.05 level of significance, does this support claim that the second production li does superior work? **PU2018**
251. A social study group wants to determine views expressed by people that there is too much violen on television in these days. It is noted that 70% of 2,000 men selected at random believed so, 60% 4,000 women selected at random also believed that there is too much violence on television. The group wants to test whether there is a significant difference in the viewers of men as compared to views expressed by women. **PU2015**
252. At a certain date in a large city 400 out of a random sample of 500 men were found to be smoke After the tax on tobacco had been heavily increased another random sample of 600 men in the sa city included 400 smokers. Was the observed decrease in the proportion of smokers significant? **PU2009**
253. A sample of 500 respondents was selected in a large metropolitan area to determine various information concerning consumer behavior. Among the question was, "Do you enjoy shopping f clothing?" of 240 males, 168 answered yes. Of 260 females, 220 answered yes.
Is there evidence of significant difference between males and females in the proportion that enjoy shopping for clothing at 0.05 level of significance? **Ans: -4.032**
254. In a factory, 2% fans were found defective in a lot of 2000 fans and in another factory, 2.5% fa were found defective in a lot of 3000 fans. Do you find that the fans in the second factory are significantly inferior compared to the fans in the first factory. Use α=5% **PU2017 1.15**
255. In a certain factory, 2% fans were found defective in a lot of 3000 fans and in another factory, 3 fans were found defective in a lot of 2500 fans. Do you find that the fans in the second factory ar significantly inferior compared to the fans in the first factory. Use α=5% **PU2017 2.38**
256. In a random sample of 1000 houses in a certain city it was found that 318 own color T.V. In the neighboring city it was found that 450 out of 1200 houses own color T.V. Test whether there is significant difference between the two samples of housing with respect to using color T.V. at 5% level of significance
257. Surveys were conducted in Kathmandu and Pokhara to ascertain viewers' habits regarding Kant Television. In Kathmandu, 1000 people were interviewed and 680 said they viewed Kantipur

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Television. In Pokhara 600 people were interviewed and 444 said they viewed Kantipur television. Is there a significant difference between viewing habits in Kathmandu and Pokhara? Support your answer at 5% significance level. **Ans: -2.54**

258. A machine puts out 20 defective articles in a sample of 400. After the machine is overhauled, it puts out 10 defective articles in a batch of 300. Has the machine improved? **Ans: 1.33**

259. A social study group wants to determine if people think that there is too much violence on television in these days. 70% of 2000 men selected at random believed so 60% of 4000 women selected at random also believed that there is too much violence on television. The group wants to test if there is significant difference in the viewers of men as compared to views expressed by women. Test the hypothesis at 0.05 **PU2016**

260. The nine items of the sample had the following values 45, 47, 50, 52, 48, 47, 49, 53, and 50. Does the mean of the nine items differ significantly from the assumed population mean of 57? **Ans 9.5**

261. Ministry of Tourism and Civil Aviation has claimed that the average length of stay of tourist in Nepal is 13 days. To test this claim a researcher asked 9 tourists about their length of stay in Nepal and their length of stay in days were 10, 15, 11, 5, 7, 4, 8, 14 and 11. On the basis of this sample result can we conclude that the average length of stay is 13 days? **Ans: -2.82**

262. A random sample of 10 boys had the following I.Q's: 70, 120, 110, 83, 88, 95, 98, 107, 99 and 100. Do these data support the assumption of a population mean I.Q. of 100? Does this data support the assumption of a population mean IQ scores of 100? **PU2018**

263. The national bureau of standard has previously reported the value of Se (Selenium) in NBS orchad leaves to be 0.08. a random sample of size n=6 gives the following determinations:**PU2013 2017**
0.072, 0.073, 0.08, 0.078, 0.088, 0.08
Does the data contradict the previously reported value?

264. The accompanying data on cube compressive strength (MPA) of concrete specimens appeared in the article "Experimental Study of Recycled Rubber Filled High Strength Concrete".
112.3 97 92.7 86 102 99.2 95.8 103.5 89 86.7
Suppose the concrete will be used for a particular application unless there is strong evidence that true average strength is less than 100 MPa. Should the concrete be used? **PU2016**

265. Ministry of Tourism and Civil Aviation has claimed that the average length of stay of tourist in Nepal is 13 days. To test this claim a researcher asked 9 tourists about their length of stay in Nepal and their length of stay in days were 10, 15, 11, 5, 7, 4, 8, 14 and 11. On the basis of this sample result can we conclude that the average length of stay is more than13 days?**Ans:-2.83**

266. The manufacturer of the Energy saver furnace claims a mean energy efficiency of at least 0.83. a sample of 21 Energy saver furnaces gives a sample mean of 0.81 and sample standard deviation of 0.06. Data show approximately a normal distribution. Test whether the manufacturer's claim can be rejected at the 5% level of significance. **PU2019**

267. The average life of 10 houses is 88 years and standard deviation is 10 years. Test the hypothesis that the average life of houses is 90 years. Use 5% level of significance.

268. For a random sample of 10 pigs fed on diet A the increase in weight (in lbs) in a certain period was 10, 17, 13, 12, 9, 8, 14, 15, 6, and 16. For another random sample of 12 pigs fed on diet B, the increase in weight in the same period were 14, 18, 8, 21, 23, 10, 17, 12, 22, 15, 7, 13. Test whether diets A and B differ significantly as regards their effect on an increase in weight**Ans: 1.5 , acceptHo**

269. The average gas mileage for 12 cars of brand A was 27.2 miles per gallon and standard deviation was 3.8 miles per gallon. The 9 brand B cars that were tested averaged 32.1 miles per gallon and the standard deviation was 4.3 miles per gallon. At 0.01 level of significance should it conclude that brand A cars have lower average gas mileage than do brand B cars? **Ans:2.628**

270. The following random samples are measurements of the heat producing capacity (in millions of calories per 1000kg) of specimens of coal from two mines**PU2014** **Ans:2.61**

Mine 1	82	81	83	80	83	
Mine 2	72	78	79	81	79	78

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Use the 0.01 level of significance to test whether the difference between the means of these two mines is significant.

271. Two independent samples of 8 and 7 items respectively had the following values, whether these samples are drawn from the populations with equal mean. Test at 5% level of significance

Sample I	9	11	13	11	15	9	2	14
Sample II	10	12	10	14	9	8	0	

272. Two horses A and B were tested according to the times (in seconds) to run a particular track with the following results.

Horse A	28	30	32	33	33	29	34
Horse B	29	30	30	24	27	27	32

Analyze the above data and report whether or not you can discriminate between the two horses at 5% level of significance. **PU2003**

273. Describe the importance of sampling. Two types of drugs were used on 5 and 7 patients for reducing their weight. Drug A was imported and drug B indigenous. The decrease in the weight a using the drugs for six months was as follows: **Ans : 0.73**

Drug A	12	13	11	14	10		
Drug B	9	12	14	15	10	9	8

Is there a significant different in the efficiency of two drugs?

274. Two sales of items in 8 shops before and after advertisement are given as:

Before	70	65	48	72	80	92	98	100
After	72	70	53	75	84	95	105	104

Test whether advertisement was effective or not at 1% level of significance.

275. Suppose a doctor wishes to test the effect of a new diet designed to reduce the blood sodium le of 10 patients. The patient's sodium levels are checked before they are placed on the diet and aft two weeks on the diet. At the 0.05 level of significance, is there enough evidence to conclude tha the diet is effective in lowering the sodium content of their blood?

Before	146	138	152	163	136	147	148	141	143	142
After	135	133	147	156	138	141	139	132	138	131

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276. BBA students were selected by a big organization and they were given an IQ test immediately a the selection and their score out of 50 were noted. After two months training was arranged for the and on compietion, IQ test was given and scores recorded.

Students	A	B	C	D	E	F	G	H	I	J
Scores (before training)	31	35	39	26	41	45	41	35	19	32
Scores (after training)	32	30	34	35	38	42	44	39	29	37

277. The sales data of an item in six shops before and after a special promotional campaign are as und

Shops	A	B	C	D	E	F
Before campaign	53	28	31	48	50	42
After campaign	58	29	30	55	56	45

Can the campaign be judged to be a success? Test at 5% level of significance.

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278. Memory capacity of 10 students was tested before and after training. State. whether the training was effective or not at 5% level of significance from the following data.

Before	12	14	11	8	7	10	3	0	5	6
After	15	16	10	7	5	12	10	2	3	10

279. Eleven students of BE were given a test in statistics. They were given 15 days special coaching given and thereafter were given second test. Marks obtained in the two tests are below.

Studens	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
I test	23	20	19	21	18	20	18	17	23	16	19
II test	24	19	22	18	20	22	20	20	23	20	18

Do the marks indicate the special coaching has benefited the students at 5% level of significance?

280. Memory capacity of 10 students was tested before and after training. State, whether the training was effective or not at 5% level of significance from the following data.

Before	12	15	12	8	10	11	15	5	14	2
After	14	18	10	15	16	8	9	10	12	14

281. 10 engineering students were selected at a big organization. They were given an IQ test immediately after selection and their scores out of 50 were noted. Two month training was arranged for them and on completion of IQ test was again given and score recorded. **PU2012** ***Ans:-0.488***

Students	1	2	3	4	5	6	7	8	9	10
Score (before)	31	35	38	25	40	45	40	38	18	30
Score (after)	32	30	32	37	38	37	42	40	27	35

On the basis of these results, can it be concluded that the training has benefited the students?

282. ABC physical fitness claims that completion of their weight loss programme will result in a weight loss. To test this claim, 6 persons were selected of random and they were put through the weight loss programme and weights before and after the programme recorded. Test the claim of fitness center at 0.05 level of significance. The weights in pounds in six persons recorded before and after the programme are as follows. **Pu2016**

Persons	1	2	3	4	5	6
Weight (before)	145	200	160	185	164	175
Weight(after)	143	190	165	183	160	176

283. The sales figures of an item in eight shops before and after advertisement is given as:

Before	70	65	48	72	80	92	98	100
After	72	70	53	75	84	95	105	104

Test whether advertisement was effective at 5% level of significance. **PU2013** ***Ans: -7.33***

284. An IQ test was administered to 7 persons before and after they were trained. The results are given below:

Candidates	I	II	III	IV	V	VI	VII
IQ before training	130	135	120	133	142	137	126
IQ after training	138	131	130	135	146	146	138

Test whether the training is designed properly. **PU2015**

285. An IQ test was administered to 5 persons before and after they were trained. The results are given below:

Candidates	I	II	III	IV	V
IQ before training	110	120	123	132	125
IQ after training	120	118	125	136	121

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Is training effective at 5% level of significance? Ans: 0.8166 PU2015

286. The quantity demanded of a certain product before advertisement and after advertisement is giv
below.

Before advertisement	50	40	45	50	60	70	35	38	60	70
After advertisement	55	41	51	51	55	75	50	54	70	78

Is the advertisement effective? Test at 5% level of significance. **PU2015**

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sement is given

Correlation and regression Analysis

287. Calculate correlation coefficient between height in inches of father (x) and their sons (y) **Ans:0.603**

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

288. The following table gives the marks in Physics and Chemistry by group of 10 students. Find the correlation coefficient and the regression equation of y on x. hence estimate the marks scored by a student in Chemistry who scored 52 marks in physics. **PU2012 Ans:-0.11**

Marks in physics	37	35	16	28	23	27	19	20	16	11
Marks in chemistry	24	29	19	14	30	19	27	30	20	30

289. The following table given the age x and blood pressure y of 10 women.

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	149	150

a. Find correlation coefficient between age and pressure. Also find coefficient of determination and interpret it. **Ans:0.8917,** $\hat{x} = -49.29 + 0.7163y$

e. Fit a regression equation of X on Y **PU2015**

290. From the following table showing age of cars of a certain make and annual maintenance costs, obtain the regression equation for cost related to age.

Age of cars in years	2	4	6	8	10	12	14
Annual maintenance cost(000)	2	5	6	7	10	11	15

Also calculate maintenance cost for the year 20 years old.

291. Scraps of iron were selected on the basis of their densities, X and their iron contents Y were measured. The results were as follows. **PU2019**

X	2.8	2.9	3	3.1	3.2	3.2	3.2	3.3	3.4
Y	27	23	30	28	30	32	34	33	30

a) Find the relationship between their densities and their iron content.

b) Find the regression equation of y on x by the method of least squares.

292. The following table gives the age of husbands and wives at the time of their marriage. **Ans:0.82**

Age of husband (X)	23	27	28	28	28	30	30	33	35	38
Age of wives (Y)	18	20	22	27	21	29	27	29	28	29

Find the correlation coefficient between x and y **PU2015**

293. Shear stress (y) and rate of shear (x) can be measured for a liquid in a viscometer. For 12 pairs of values the data can be summarized as **PU2016**

$\sum X=132, \sum Y=151.7, \sum X^2=1944, \sum Y^2=2570.48, \sum XY=2233.2$

a. Find the linear regression equation of y on x

b. What is the correlation coefficient?

c. What fraction of the variance of y is explained by x?

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294. A study reported in the article "the effects of Water Vapour Concentrations on the Rate of Combustion of an Artificial in Humid Air Flow" (Combustion and Flame) give data on

X=temperature of nitrogen oxygen mixture (1000 degree F) under the specified conditions

Y= oxygen diffusivity **PU2015**

Summary quantities are:

n=9

$\sum X=12.6, \sum Y=27.68, \sum X^2=18.24, \sum Y^2=93.344, \sum XY=40.968$

a. Calculate correlation coefficient between X and Y and interpret it

b. Fit a regression equation of Y on X

c. Find correlation of determination and interpret.

295. In the United States, during the fall harvest season Pumpkin are sold in the large quantities at the farm stands. Often instead of weighing the Pumpkin prior to sale, the farm stand operators will just places the Pumpkin in the appropriate circular cutout on the counter. When asked why this was done

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one farmer replied, `` I can tell the weight of the pumpkin from its circumference``. To determine whether this was really true; a sample of 8 Pumpkin's circumference and weights were measured:PU2016

Circumference (cm)	50	55	54	52	37	52	53	47
Weight (gram)00	12	20	15	17	50	10	15	14

a) Develop the estimating linear regression equation to predict the weight of the pumpkin that is 6 cm in circumference.
b) Calculate correlation coefficient between circumference and weight of the pumpkin and interpret it

296. An instructor is interested in finding out the relationship between the number of students absent in school (y) and the day temperature (x). A random sample of 10 days was used for the study. The following table gives the record of the study.PU2014

Temperature (x)	10	20	25	30	40	45	50	55	59	60
No of absent (y)	8	7	5	4	2	3	5	6	8	9

Find the regression equation to estimate the number of students absent in the school when the temperature is 65.

297. Calculate the coefficient of correlation from the following marks obtained by 10 students in mathematics (x) and statistics (y) PU

X	40	65	60	25	85	35	45	70	80	55
Y	30	85	65	35	90	35	55	75	75	45

Also obtain the regression equation of y on x

298. The following are the heights in centimeter and the weights in kilogram of 10 men:

Height	162	168	174	176	180	180	182	184	186	186
Weight	65	65	84	63	75	76	82	65	80	81

a. Develop the estimating regression equation of weight on height
b. Find the coefficient of determination and interpret it. Also find standard error of the estimate.PU2017

299. An engineering student has a summer job with the forestry service. He measured the tree trunk diameters (x in inches) and related them to the age of the tree (y in years) The following information was obtained:
$n= 6, \sum X=21, \sum Y=26, \sum X^2=91, \sum Y^2=142.52, \sum XY=113.8$ PU2018

a) Find the regression equation of y on x and estimate the age of the tree whose diameter is 4 inches.
b) Find the correlation coefficient between x and y and interpret its meaning.
c) Determine coefficient of determination and interpret it.
d) Compute standard error of the estimate.

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300. Represent by means of histogram, frequency polygon and frequency curve PU2005

Wage	10-15	15-20	20-25	25-30	30-40	40-60	60-80
No of workers	7	19	28	15	12	12	8

301. Represent the following data by means of histogram, frequency curve and polygon

Salaries	300-310	310 320	320-330	330-350	350-370	370-400
No of workers	7	19	28	15	12	12

302. The observed values are as follows **PU2009**

5.1, 14.4, 14.7, 10.8, 6.5, 5.7, 7.7, 14.1, 9.5, 3.7, 8.9, 7.9, 7.9, 4.5, 10.1, 5.0, 9.6, 5.5, 5.1, 11.4, 8.0, 12.1, 7.5, 8.5, 13.1, 6.4, 18.0, 27, 18.9, 10.8, 13.1, 8.4, 16.9, 2.7, 9.6, 4.5, 12.4, 5.5, 12.7, 17.1

a. Construct a frequency distribution for the data set using 0.0-4.0 and 4.0-8.0 as the first and second class interval respectively.
b. Draw the histogram and frequency curve of the above frequency distribution.

303. The daily wages of workers of a factory are given below: **PU2014**

Wages (Rs)	300-310	310-320	320-330	330-350	350-370	370-410
No of workers	8	10	20	18	16	12

a. Construct a histogram and frequency polygon for the data.
b. Draw an ogive for the data and estimate the median age.

304. The weight in lbs of 40 boys in a class are as follows: **PU2013**

138	172	145	147	150	119	158	152	168	142
157	147	102	144	165	136	164	163	128	135
126	150	146	148	145	125	146	153	138	156
173	140	135	149	140	144	132	154	142	135

a. Construct a frequency distribution
b. Draw a less than ogive and find number of boys whose weight is less than 165 lbs

305. The weight (in lbs) of 40 boys in a class are as follows:

138	172	145	147	150	119	158	152	168	142
157	147	102	144	165	136	164	163	128	135
126	150	146	148	145	125	146	153	138	156
173	140	135	149	140	144	132	154	142	135

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a. Construct a frequency distribution.
b. Construct the histogram and frequency curve.

306. The marks obtained by 50 students in an Examination are given below:

25, 30, 67, 84, 28, 32, 65, 70, 88, 43, 54, 49, 52, 42, 42, 52, 51, 58, 59, 50, 44, 38, 61, 76, 87, 41, 42, 50, 50, 56, 58, 37, 30, 35, 36, 35, 63, 65, 60, 49, 35, 54, 45, 59, 40, 44, 52, 51, 73, 61

i) Classify the above data taking class interval 20-30, 30-40, etc
ii) Draw the histogram and frequency curve of the above frequency distribution.

307. Following data represents the tensile strength of steel rod manufactured by company a located at Biratnagar **PU2017**

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65	36	49	84	79	56	28	43	67	36
43	78	37	40	68	72	70	55	62	82
88	50	60	56	57	46	39	57	22	65
59	48	76	74	80	69	51	40	56	45
35	21	62	52	63	32	86	64	53	34

Construct a frequency distribution and represent the data by means of Cumulative frequency curve. Identify the median and first quartile from the curve. Also interpret the result of first quartile.

308. From the following frequency distribution **PU2016**

Income (Rs.000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of employees	5	10	18	23	7	6

Construct an ogive that will help you answer; find the number of employees having income:

i) Less than Rs. 35,000
ii) Between Rs. 20,000 and Rs. 50,000
iii) More than Rs. 25,000

309. The following table shows the length of eighty bally bridges: **PU2015**

68	84	73	82	68	90	62	88	76	93
73	79	75	73	60	93	71	59	85	75
61	65	88	87	74	62	95	78	63	72
66	78	75	75	94	77	69	74	68	60
96	78	82	61	75	95	60	79	83	71
79	62	89	97	78	85	76	65	71	75
65	80	67	57	88	78	62	76	53	74
86	67	73	81	72	63	76	75	85	77

a. Construct a grouped frequency distribution having class width 10
b. Draw a less than ogive and more than ogive in same graph and hence locate median
c. By the help of less than ogive find the number of bridge having length less than 65 meter

310. After the implementation of an economic program to uplift the economic condition of a commun following information were found. **PU201**

Monthly income (Rs000)	4-6	6-8	8-10	10-12	12-14	14-16	16-18
After the plan (no of families)	8	65	37	15	15	5	5

Construct an ogive to find

I) The number of families whose monthly income is between Rs. 8,000 to Rs.14, 000.
II) Find the number of families whose monthly income is above Rs. 12,000

311. A national association of the Real State sellers has collected these data on a sample of 1 salespeople representing their total commission earning annually:

Commission ($000)	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
Frequency	6	12	15	35	40	23	11	8

Construct an ogive that will help you answer these questions:

i) About what proportion of the salespeople earns less than $18,000?
ii) How many people earns in between 15000 to 35000?

312. From the following distribution of mark of 500 students of a college, find the minimum pass mark if only 20% of student had failed and also the minimum mark obtained by the top 25% of the students. Represent the data by histogram. Ans: 30, 58.3 **PU 2017**

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Marks	0-20	20-40	40-50	50-60	60-80	80-100
No. of students	50	100	150	90	60	50

313. The table given below represents the daily wage distribution of 130 workers. Find out the range of income of the middle 60% workers **PU2015 Ans: 102.5, 140, 129.786, 105.76**

Wage more than	70	85	100	115	130	140	160	175
No of workers	130	120	109	79	44	26	14	5

a. Find the minimum income of top 35% workers?
b. Find the maximum income of initial 25% workers?

314. The test scores of the students in probability and statistics are listed below. Construct a stem and leaf display

92	78	73	89	98	89	83	75	83	94
99	69	71	96	67	81	73	88	86	82
63	73	76	82	84	89	92	95	78	87

Also find the lowest score of the best 25% of the students

315. Given below the pretax monthly income of the workers of FEWA hydropower in 2000

Income more than (000)	7	6	5	4	3	2	1	0
No of workers	2	8	10	15	25	40	55	60

Draw a less than ogive and hence find out
a. The highest income of the lowest 50% of the worker and
b. The minimum income earned by the top 5% of the workers **PU2012**

316. A random sample was taken of the thickness of insulation in transformer windings, and the following thickness in mm were recorded:

18	21	22	29	25	31	37	38	41	39
44	48	54	56	56	57	47	38	35	36
29	37	32	42	43	40	48	36	37	37

a. Prepare a stem and leaf display
b. Prepare a box and whisker plot

317. Over a period of 40 days the percentage relative humidity in a vegetable storage building was measured. Mean daily values were recorded as shown below:**PU 2017**

60	63	64	71	67	73	79	80	83	81
86	90	96	98	98	99	89	80	77	78
71	79	74	84	85	82	90	78	79	79
78	80	82	83	86	81	80	76	66	74

a) Prepare a stem and leaf display for these data. Show the leaves sorted in order of increasing magnitude on each stem.**Ans: 60 74.5 80 84.75 99**
b) Draw a box plot for these data and interpret the data in practical manner.

318. The following are the numbers of minutes that a person had to wait for the bus to work on 15 working days
10, 1, 13, 9, 5, 9, 2, 10, 3, 8, 6, 17, 2, 10, 15

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Draw a box and whisker plot and interpret it. PU2014.

319. The following data set represents the number of new computer accounts registered during ten consecutive days. PU2018

43, 37, 50, 51, 58, 105, 52, 45, 45, 10.

I) Compute the mean, median and standard deviation.

II) Draw a box and whisker plot and and identify whether it is skewed or not

320. An investigator wants to study the speed of cars at Araniko highway and he collected the speed of 30 vehicles and speed were,

35 37 42 45 47 48 50 55
67 70 75 80 90 95 94 48
55 60 71 63 70 65 80 55
40 35 36 85 79 30

a) Present the above data in stem and leaf display.

b) Construct continuous frequency distribution using Struge's rule and construct the cumulati curve and find median speed, speed of first 25% vehicles, speed of first 75% vehicles and al compute the percentage of vehicles whose speed lies between 40 to 70 km.

321. In order to estimate how much water will be needed to supply for Baneshwor area in Kathmand valley in the next year, the town council asked to find out how much water a sample of famil currently uses. The sample of 15 families used the following number of gallons (in thousands) in t past year:

11.2	21.5	16.4	19.7	14.6
16.9	32.2	18.2	13.1	23.8
18.3	15.5	18.8	22.7	14.0

i) List the five number summaries.

ii) Form the Box and Whisker plot, determine the shape and explain whether the use of the water likely to be high or low? **Ans:11.2, 14.6, 18.2, 21.5, 32.2**

322. In Kathmandu saving banks are permitted to sell a form of life insurance called saving bank l insurance. The approval process consists of under writing, which includes a review of t application, a medical information bureau check, possible request for additional medical informati and medical exams and a policy compilation state where the policy pages are generated and send the bank for delivery. The ability to deliver approved policies was selected and the total processi time in days was recorded in the following reults.

73 19 16 64 28 28 31 90 60

56 22 18 45 48 17 17 17 91
92 63 50 51 31 56 69 16 17

i) Construct a box and whisker plot.

ii) Are the data skewed?

323. A bank is studying ATM use in Kathmandu. a sample of 30 ATM's showed they were used t following number of times yesterday. Develop a stem and leaf display chart. What was t minimum and maximum number of times each ATM was used? Also construct a histogram.

83 64 84 76 84 54 75 59 70 61
63 80 84 73 68 52 65 90 52 77
95 36 78 61 59 84 95 47 87 60

324. The following data represents the grade point average of students in a business statistics PU2015

3.4	2.1	2.4	2.8	3.2	3.7	2.5	2.2
2.8	2.6	3.2	3.3	3.3	2.9	2.4	2.5
2.4	3.2	2.3	2.8	2.8	3.1	3.1	3

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a. Construct a stem and leaf display
b. What is the GPA of most of the students in this subject?
c. Construct histogram

325. The following is a steam and leaf display for a sample of 22.

9	7	1	4				
10	8	2	2	3	0		
11	5	6	1	7	7	6	7
12	3	9	4	2	8	2	
13	2	0					

a. Rearrange the leaves and form the revised stem and leaf display.
b. Place the data into an ordered array.
c. Which of these two displays seems to give more information?

326. The weights of 11 orange picked at random from a basket are as follows:
45, 55, 30, 75, 40, 60, 65, 70, 85, 80, 75
a. List the five number summary
b. From the box whisker plot

327. The file contains data on the total fat, in grams per serving, for a sample of 20 chicken sandwiches from fast-food chains. The data are as follows:
7 8 4 5 16 20 20 24 19 30 23 30 25 19 29 29 30 30 40 56
i) Place the data into an ordered array.
ii) Construct a steam and leaf display.

328. The production manager of a press determines the average time needed to photograph one printing plate. Using a stopwatch and observing the plate markers, he collects the following times (in seconds):

20.4	20.0	22.2	23.8	21.3	25.1	21.2	22.9	24.2	24.3
22.0	24.7	25.7	24.9	22.7	24.4	24.3	23.6	23.2	21.0

Construct a box and whisker plot for the given data.

329. The lives of two models (A and B) of refrigerators in a recent survey are shown below:**PU2014**
Ans:54.94 36.2 model B

Life (No of years)	No of refrigerators	
	Model A	Model B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

a. What is the average life of each model of these refrigerators?
b. Which model has greater uniformity?

330. The temperature in a chemical reactor was measured every half an hour under the same conditions. The results were 78.1^{0}C, 79.2^{0}C, 78.9^{0}C, 80.2^{0}C, 78.3^{0}C, 78.8^{0}C, and 79.4^{0}C. Calculate the mean, median, lower quartile, upper quartile, standard deviation and coefficient of variation.**Ans: 78.98, 78.9, 78.3, 79.4, 0.653, 0.83% PU2017**

331. An analysis of monthly wages paid to the workers in two firms A and B belonging to the same industry gives the following results.**PU2009 , 2016 Ans: firm B, 4.8. 5.71, firm B, 180, 11.01**

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	Firm A	Firm B
Number of worker	500	600
Average monthly wages in $	186	175
Variance of distribution of wage	81	100

a. Which firm, A or B has large wage bill?
b. In which firm, A or B is there greater variability in wage?
c. Calculate combined mean and combined variance of wage of firm A and firm B?

332. For acceleration due to gravity, two students obtained the following value in m/s^2.
Student (I):- 9.71, 9.6, 10.2, 9.82, 10, 9.62, 9.5, 9.62, 8.9
Student (II):- 9.9, 8.9, 10.1, 9.5, 9.85, 9.4, 10.1, 9.4, 9.1
Which data are more uniform?

333. Calculate CV

Duration in seconds	No of cells
0-30	9
30-60	17
60-90	43
90-120	82
120-150	81
150-180	44
180-210	24

334. Compare the variability of the two types of bulbs

Length of life in hours	No of bulbs A	No of bulb B
500-700	5	4
700-900	11	30
900-1100	26	12
1100-1300	10	8
1300-1500	8	6

335. Kathmandu Appliance Company has just completed a study of three possible assembly line configurations for producing its bestselling two slice toasters. Configuration I has yielded a mean time to construct a toaster of 34.8 minutes, and a standard deviation of 4.8 minutes. Configuration II has yielded a mean of 25.5 minutes, and a standard deviation of 7.5 minutes. Configuration III has yielded a mean of 37.5 minutes, and a variance of 14.44 minutes. Which assembly line configuration has the least relative variation in the time it takes to construct a toaster? Which configuration will be recommendable?

336. The following are scores (runs) of two batsmen A and B in a series of 10 innings:

A	11	64	75	0	64	80	71	40	125	60
B	60	58	62	45	85	60	68	47	55	70

Who is better run getter and who is more consistent (uniform)? **Ans:65.69, 19.433, B is uniform**

337. J.J Thomson discovered the electron by isolating negatively charged particles for which he could measure the mass charge ratio. This ratio appeared to be constant over a wide range of experimental conditions and consequently, could be characteristics of new particles. His observations from two different cathode ray tubes that used air as the gas: **Ans:18.632, 10.775 tube II**

Tube I	0.57	0.34	0.43	0.32	0.48	0.4	0.4
Tube II	0.53	0.47	0.47	0.51	0.63	0.61	0.48

Which tube is more consistent and why?

338. The Pokhara University, School of Business has decided to test three new kinds of light bulbs. They have three identical rooms to use in the experiment. Bulb I has an average life time of 1470 hours and a variance of 156.Bulb II has an average lifetime of 1400 hours and a variance of 81. Bulb

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III has an average lifetime of 1350 hours and a standard deviation of 6 hours. Rank the bulbs in terms of relative variability. Which was the best bulb? Explain.

339. The means of two samples of sizes 50 and 100 respectively are 54.1 and 50.3 and the standard deviation is 8 and 7. Find the mean and standard deviation of the sample obtained by combining the two samples. Also calculate the coefficient of variation of the combined set.

340. A telefilm production company of Nepal Television is selecting a group of artist. The age in years of 20 people who were interviewed are: 50, 56, 55, 49, 52, 57, 56, 57, 56, 59, 54, 55, 61, 60, 51, 59, 62, 52, 54, and 49. The director of the telefilm wants artists whose average ages are not more than 55 years with standard deviation of at most 3 years. Does this group qualify the requirement of the director?

341. The weight of the players of Baltimore Bullets professional football team have a mean of 224 pounds with a standard deviation of 18 pounds while the mean and standard deviation of weight of their Sunday Opponent the Chicago Trailblazers is 195 and 12 respectively. Which team exhibits the greater relative dispersion in weight?PU2015

342. Mean and standard deviation of 200 items are found to be 60 and 20. If at the time of calculations two items were wrongly taken as 3 and 67 instead of 13 and 17, find the correct mean and standard deviation.

343. The means of two samples of sizes 50 and 100 respectively are 54.1 and 50.3 and the standard deviation is 8 and 7. Find the mean and standard deviation of the sample obtained by combining the two samples

344. The combined mean and variance of the salary of 250 workers of city A and city B are 560 and 5497 respectively. The mean and the variance of the salary of 100 workers of city A are 650 and 121 respectively. Find the variance of the salary in city B.

345. The mean of 100 items was 40 and the standard deviation 10 It was later discovere that two items were wrongly read as 30 and 70 instead of 3 and 27. Find the correct mean and standard deviation.

346. Find the missing information **PU2012Ans:60, 120, 8**

	Group I	Group II	Group III	Combined
Number	50	?	90	200
Standard deviation	6	7	?	7.746
Mean	113	?	115	116

347. For a computer controlled lathe whose performance was below par, workers record the following causes and their frequencies: PU2016

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Power fluctuation	6
Controller not stable	22
Operator error	13
Worn tool not replaced	2
Other	5

Construct a pareto chart

a. What percentage of the cases is due to an unstable controller?

b. What percentage of the cause is due to either unstable controller or operator error?

348. In order to improve efficiency, a manufacturer of refrigerators reviews the customer complaints within warranty period that their technicians had to attend to last year. The type of complaints and the number of complaints received are summarised below:

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Complaint type	Numbers received
Doors not aligned	15
Water dripping on floor	27
Freezer not cold enough	35
Handle trim loose	11
Shelf trim loose	9
Water line screw loose	7
Paint scratch	3
Compressor noisy	5

Construct a Pareto chart on the graph paper provided – the line curve does not have to start from the top of the first bar.

349. An article published in The Wall Street Journal discussed the costs companies face in defending their networks from cyber-attacks. The following table provides the breakdown of costs:

Cost	Percentage (%)
Consulting	11
Hardware tools	7
Labour	27
Lost business/revenue	21
Non-productive employee time	16
Software tools	13
Other	5

Construct a Pareto chart on the graph paper provided – the line curve does not have to start from the top of the first bar.

The end

Best of Luck

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